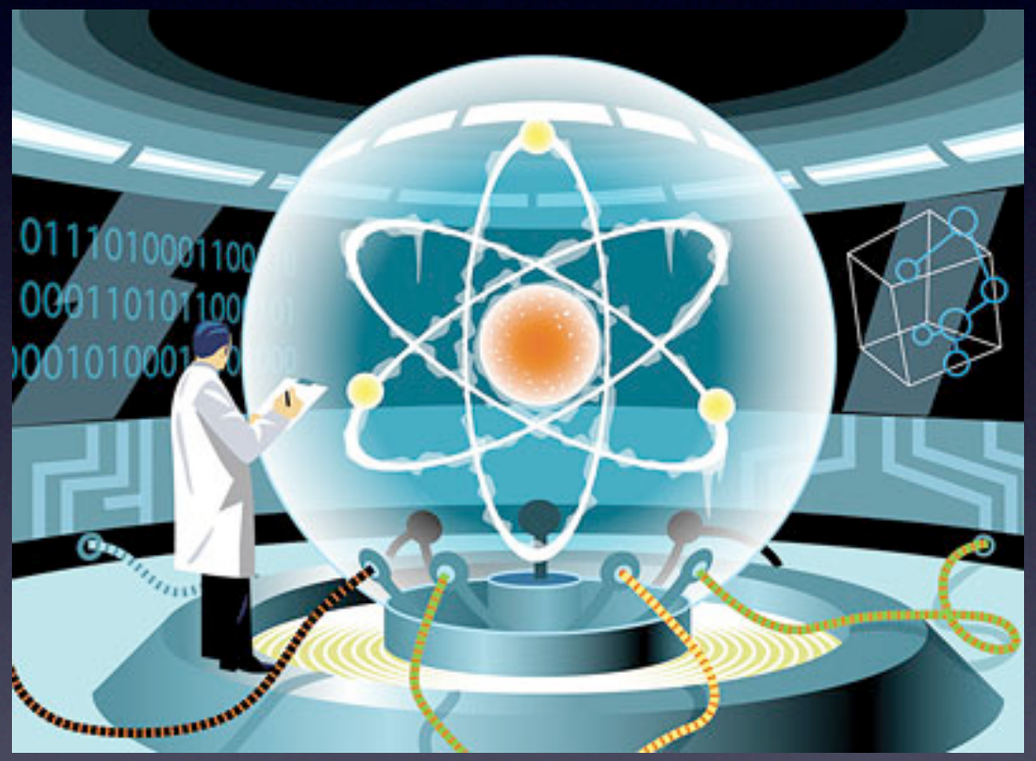
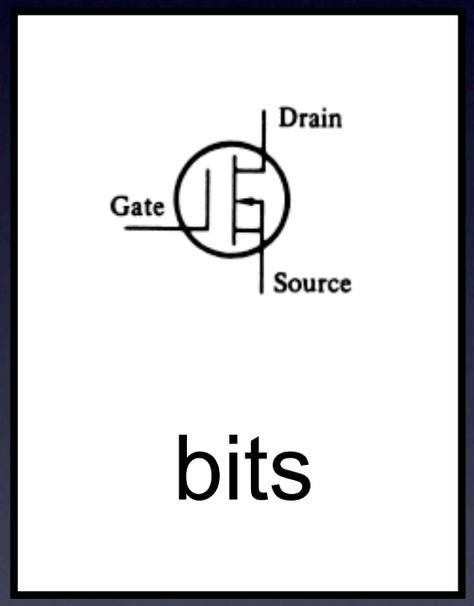
Quantum Computing Architectures



Dave Bacon
University of Washington

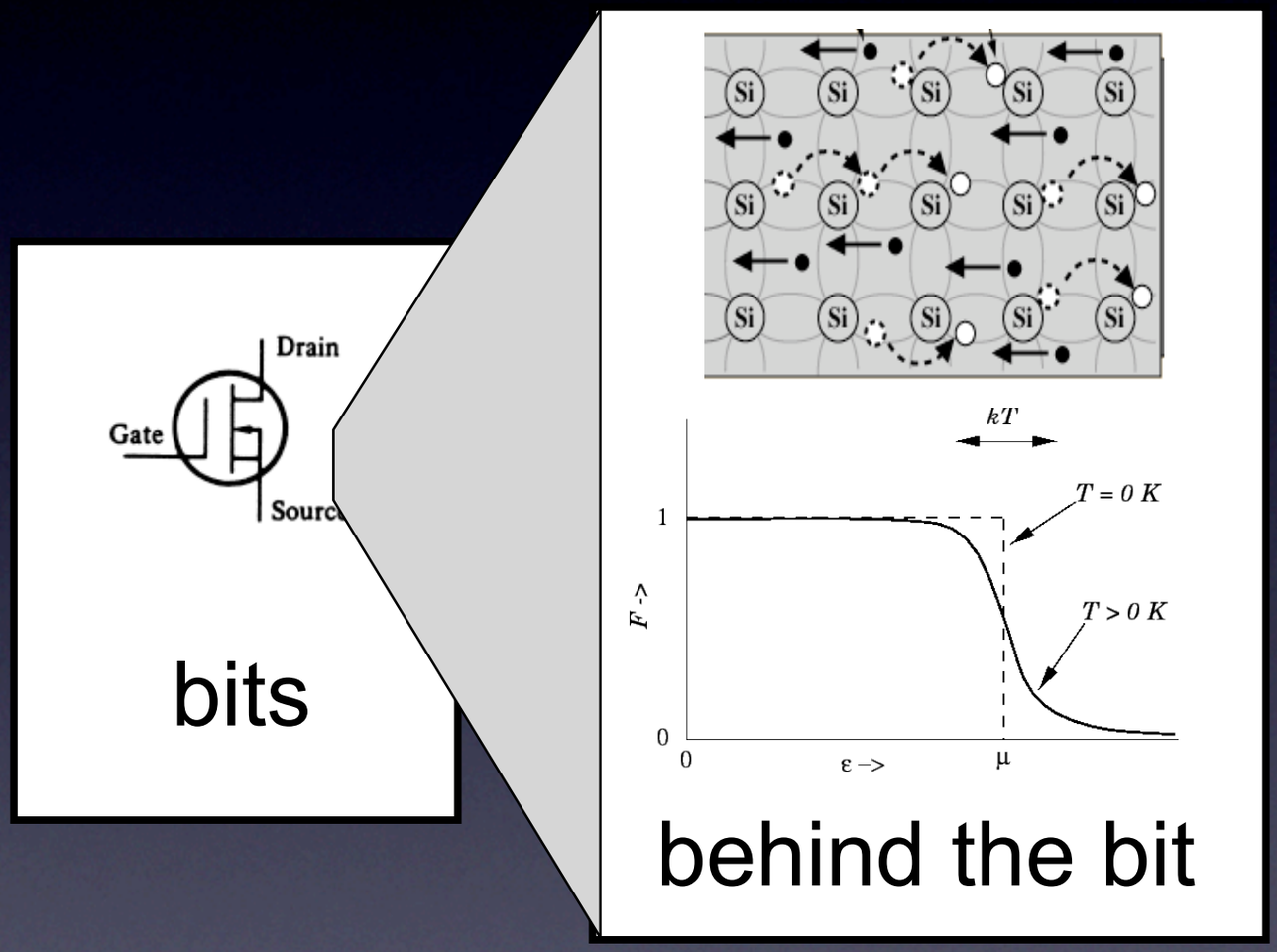
A Simple Question

Why is computation physically possible?



A Simple Question

Why is computation physically possible?



Category Errors

Analog / Digital



vs



0.9982131123....

0

Category Errors

Analog / Digital



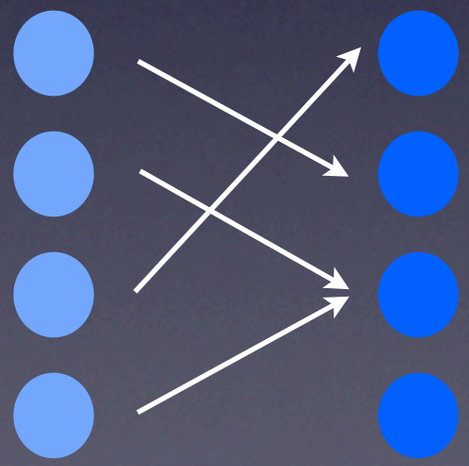
vs



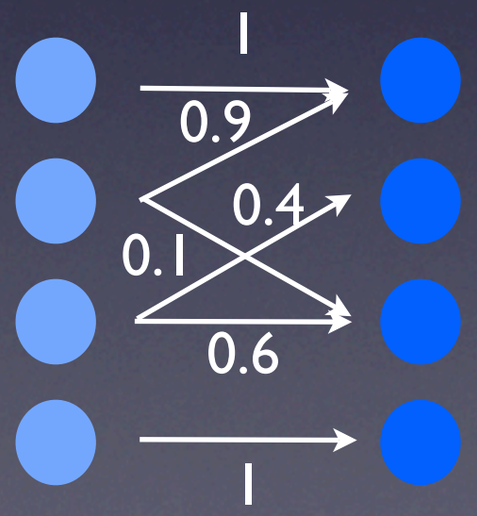
0.9982131123....

0

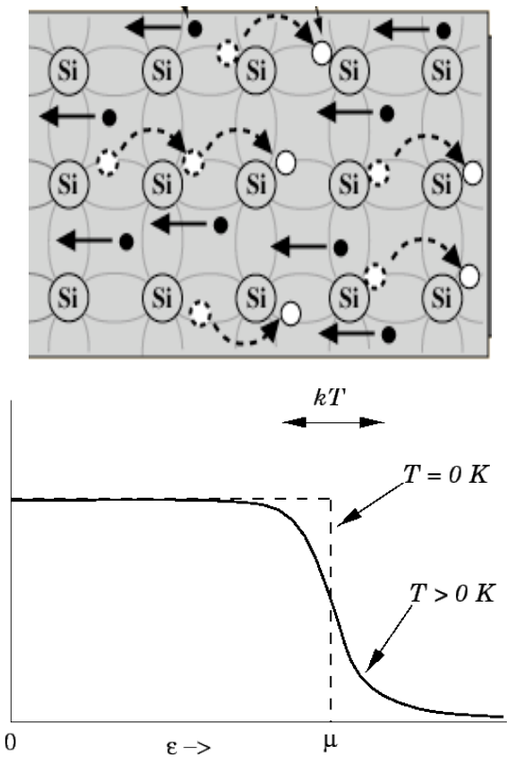
Deterministic / Probabilistic



vs



Analog Probabilistic
Digital Deterministic



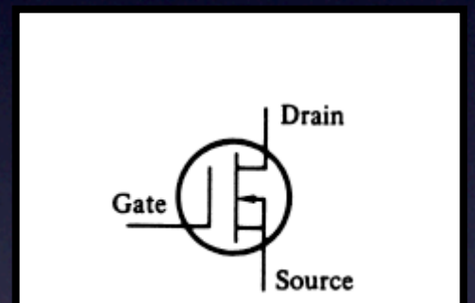
behind the bit

Analog
Voltages

Probabilistic
Electrons

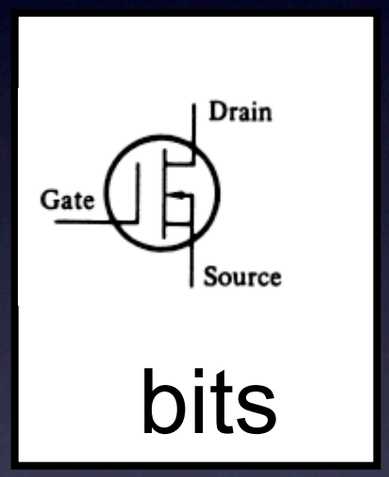
Digital
Signal

Deterministic
Gates



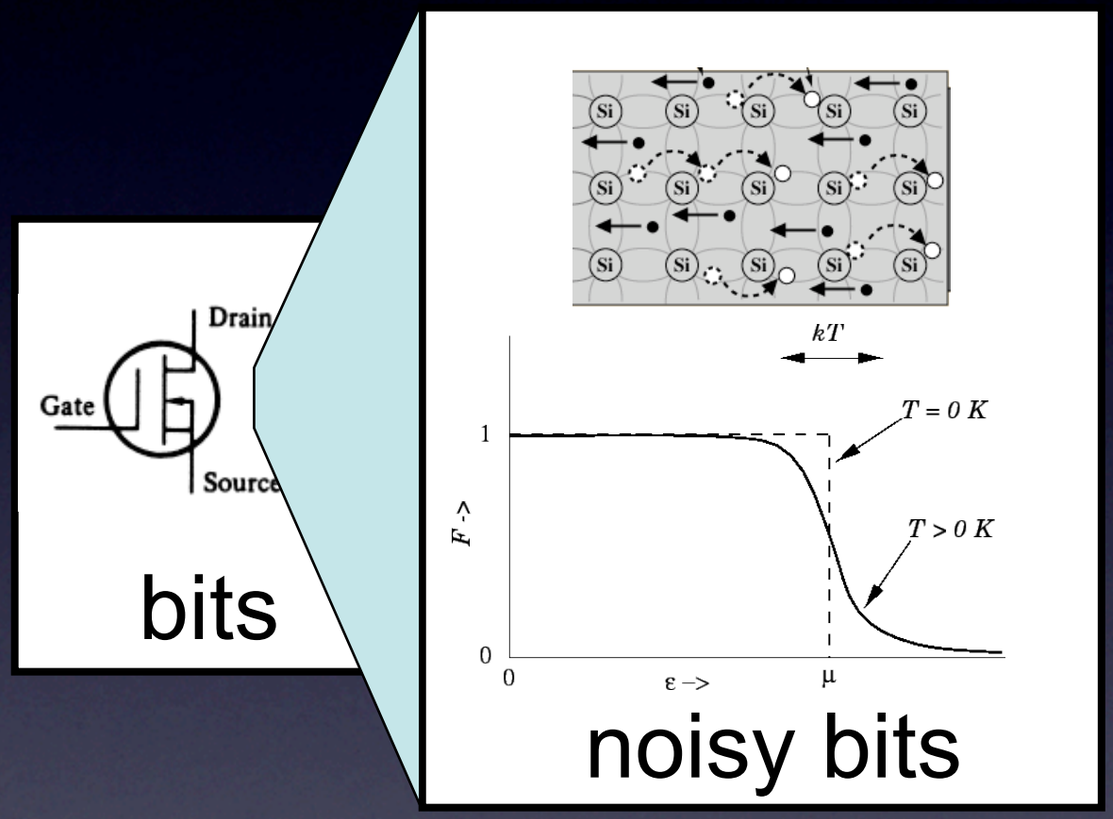
bits

Deeper



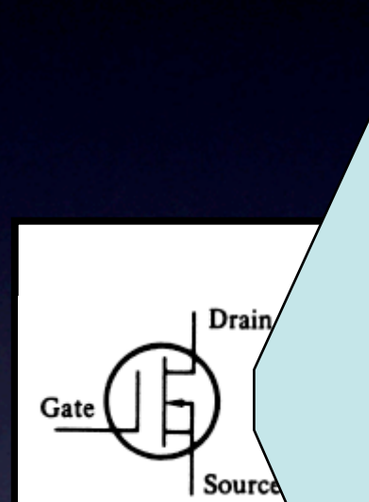
Deterministic / Probabilistic / Quantum

Deeper

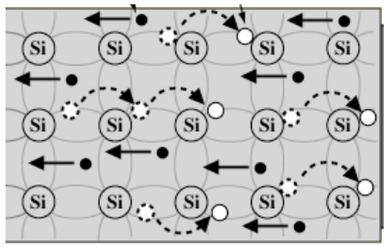


Deterministic / Probabilistic / Quantum

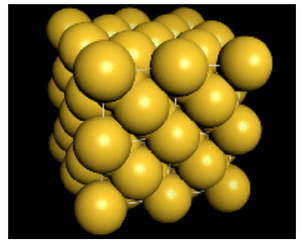
Deeper

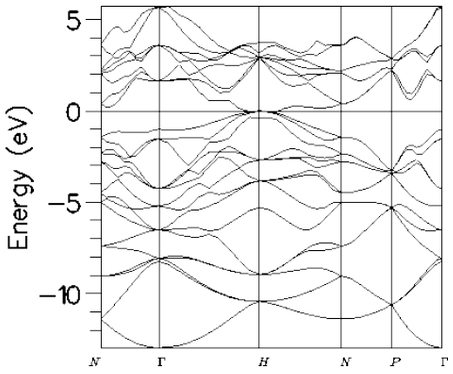


bits



noisy bits

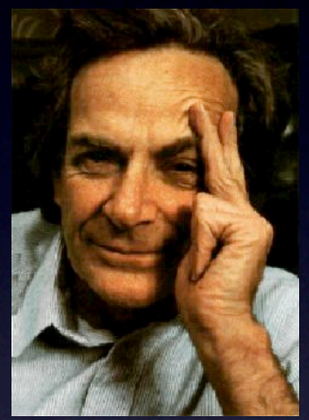




quantum bit

Deterministic / Probabilistic / Quantum

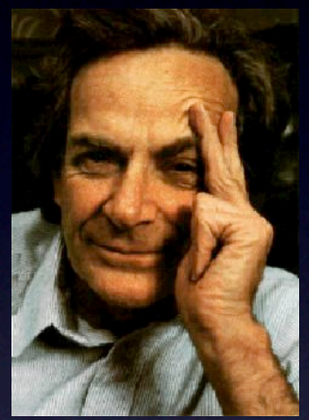
Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965

Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965



Anyone who is not shocked by quantum theory has not understood it.

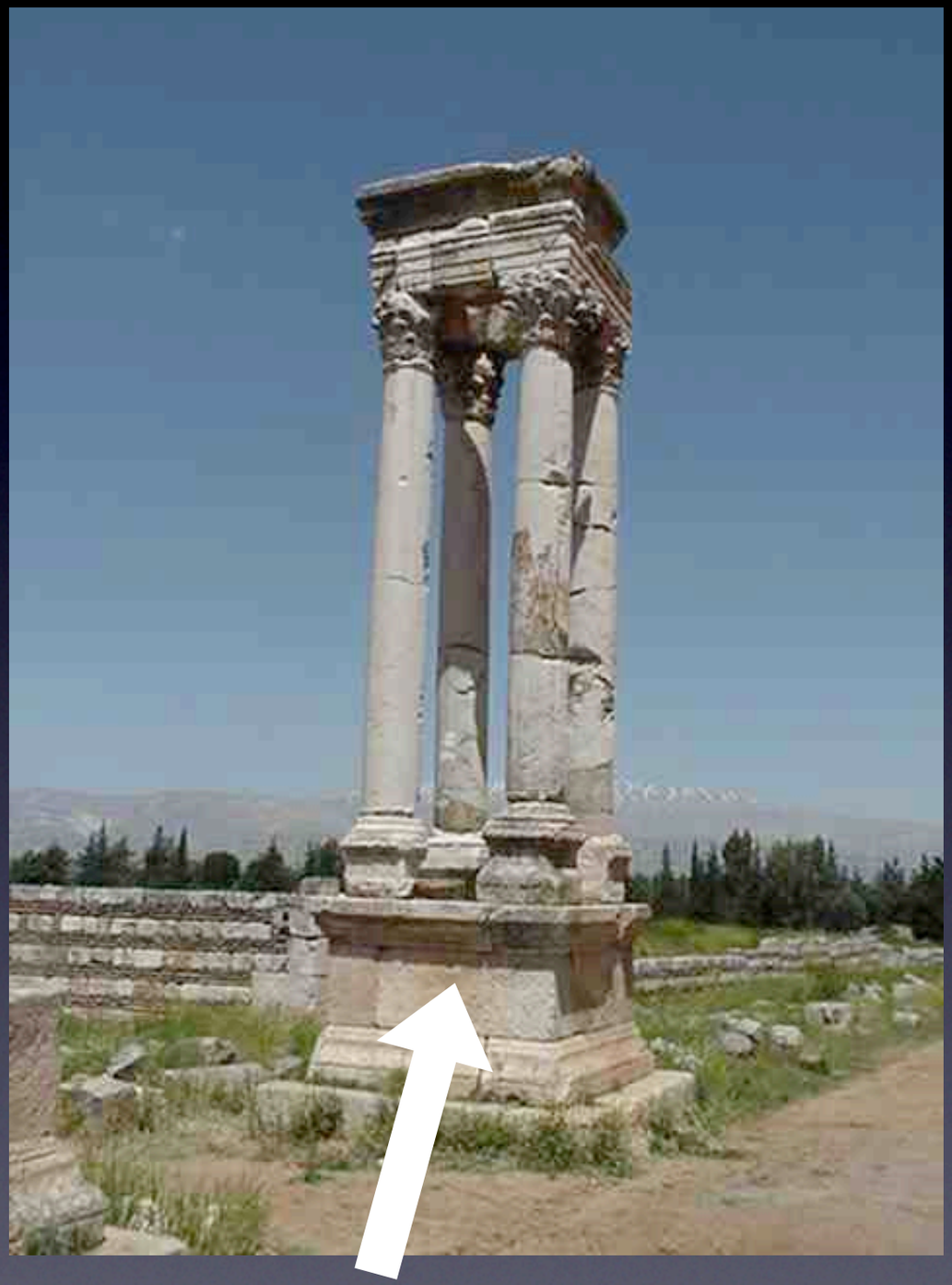
Niels Bohr
Nobel Prize 1922

Hogwash



- They understand it
- They use it
- But do they believe it?

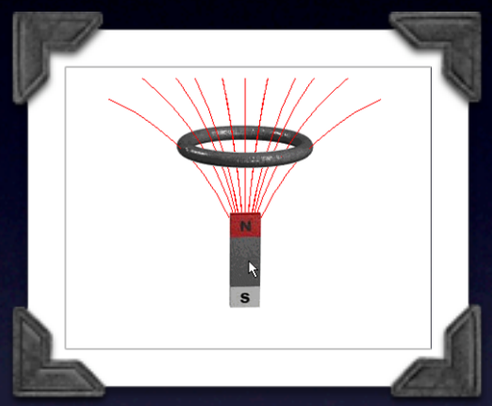
What is Quantum Theory?



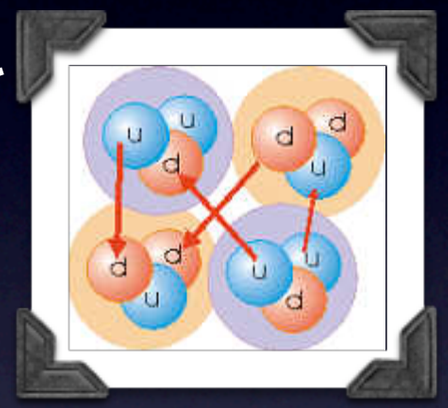
Quantum Theory

What is Quantum Theory?

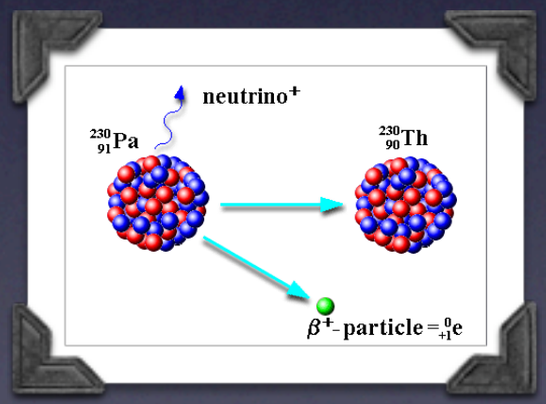
Electromagnetism



Strong force

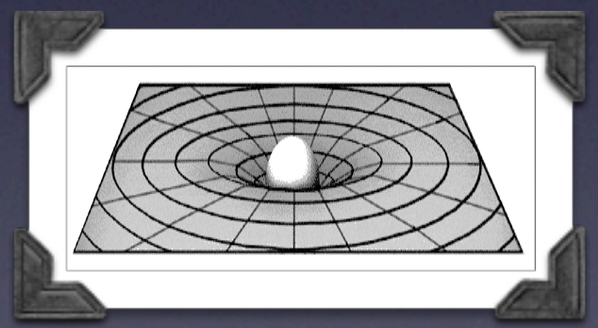


Weak force





Gravity ?!?!



Quantum Theory

Bits, Pbits, Qubits

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:			
Observation:			
Evolution:			

Bits, Pbits, Qubits

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	
Observation:			
Evolution:			

Bits, Pbits, Qubits

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
		$p \in \mathbb{R}$	$\alpha, \beta \in \mathbb{C}$
Observation:			
Evolution:			

Bits, Pbits, Qubits

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
		$p \in \mathbb{R}$	$\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent	
		1 *1-p* percent	
Evolution:			

Bits, Pbits, Qubits

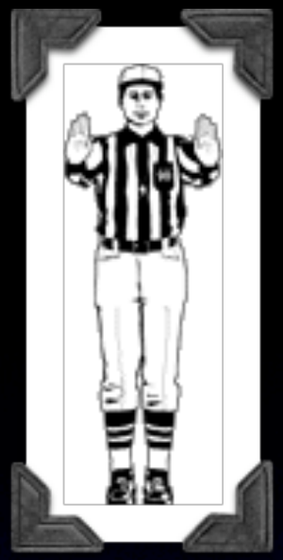
	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
		$p \in \mathbb{R}$	$\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent	0 $\lvert\alpha\rvert^2$percent
		1 *1-p* percent	1 $\lvert\beta\rvert^2$percent
Evolution:			

Bits, Pbits, Qubits

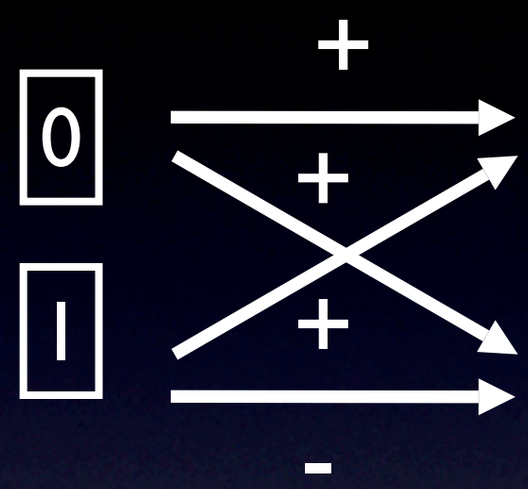
	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
		$p \in \mathbb{R}$	$\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent	0 $\lvert\alpha\rvert^2$percent
		1 *1-p* percent	1 $\lvert\beta\rvert^2$percent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1-q & r \\ q & 1-r \end{bmatrix}$	
	deterministic	stochastic probabilistic	

Bits, Pbits, Qubits

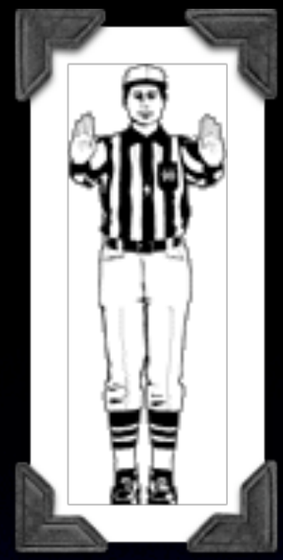
	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
		$p \in \mathbb{R}$	$\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent	0 $\lvert\alpha\rvert^2$percent
		1 *1-p* percent	1 $\lvert\beta\rvert^2$percent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1-q & r \\ q & 1-r \end{bmatrix}$	$\begin{bmatrix} u & v \\ w & x \end{bmatrix}$
	deterministic	stochastic probabilistic	unitary



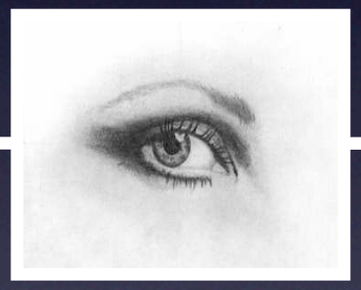
Interference



$$\begin{matrix} & 0 & 1 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{matrix}$$



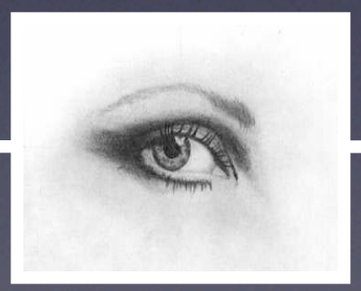
$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$



0 50%

1 50%

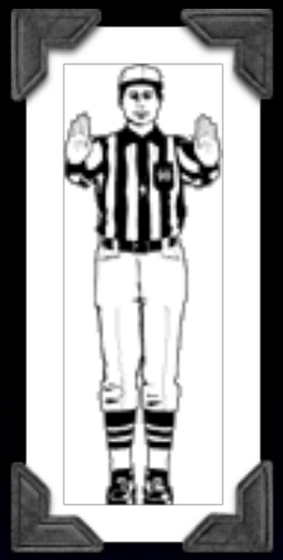
$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$



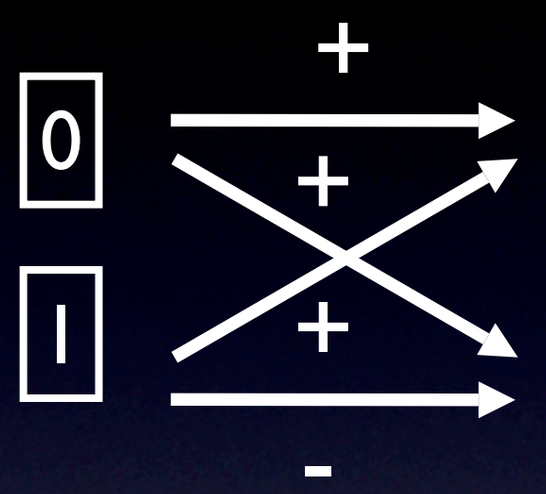
0 50%

1 50%

measure

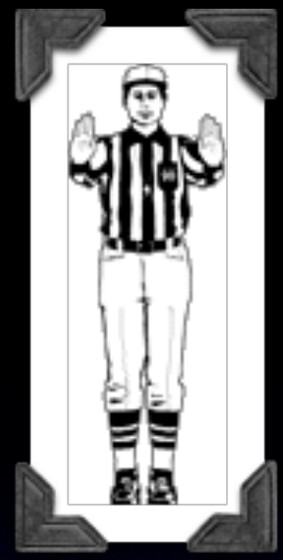


Interference

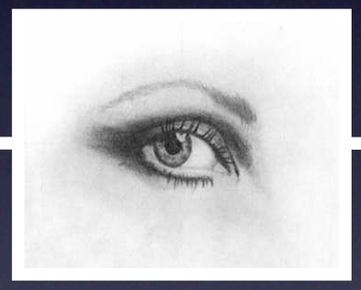


$$\begin{matrix} 0 & 1 \end{matrix}$$

$$\begin{matrix} 0 \\ 1 \end{matrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$



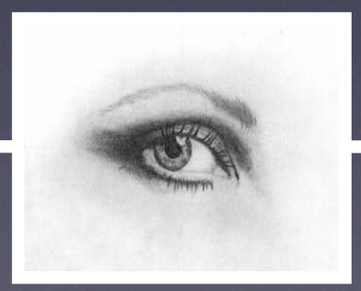
$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \quad \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$



0 100%

1 0%

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \quad \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

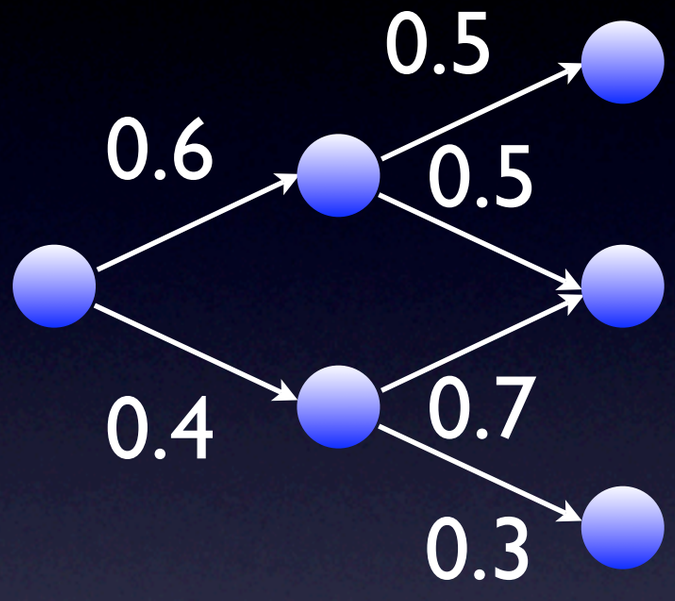


0 0%

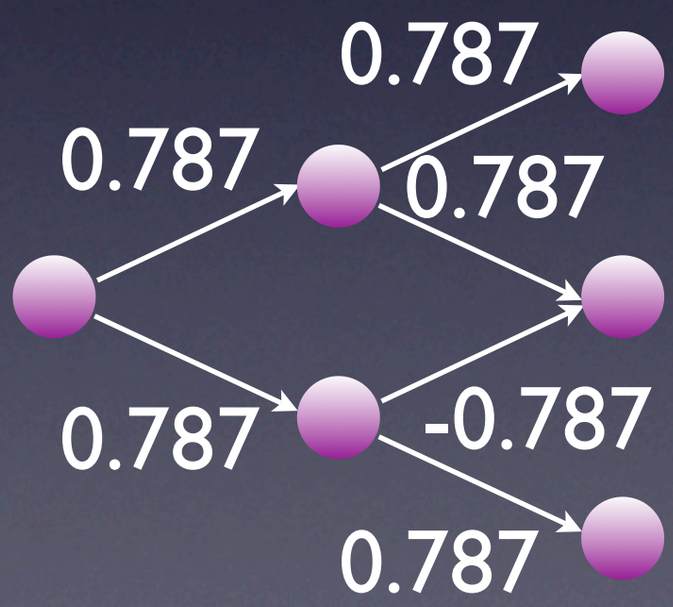
1 100%

evolve measure

Garden of Re-forking Paths



$(0.6)(0.5)+(0.4)(0.7)=0.58$



$|(0.787)(0.787)+(-0.787)\ (0.787)|^2=0$

Quantum Computers

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

x

Quantum Computers

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

3980750864240649373971255005503864911990643623425267084063851895759463889572617685833172727214610743530253622307197304822463291469530209711645985217113052071125636359039752



Peter Shor, 1994:

- Quantum computers can efficiently factor
- No known classical efficient algorithm.
- Breaks RSA, Diffie-Hellman cryptosystems.

Stop Worrying and Love Quantum Theory



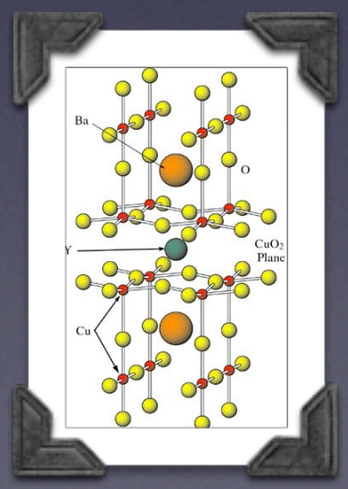
Quantum Algorithms

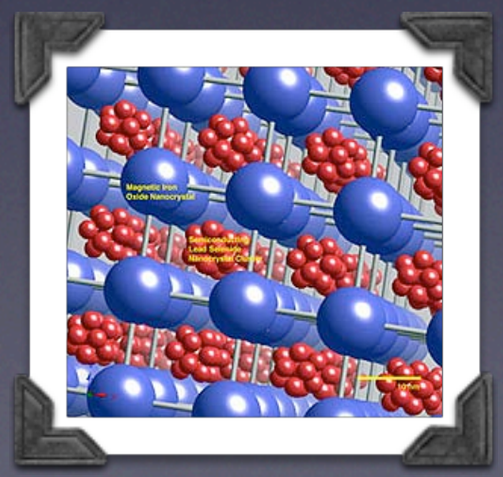
Exponential speedup

Shor's algorithm
- Breaks RSA, Diffie-Helman
- Only a few alternatives

which are note broken by quantum computers

Simulation of quantum physics





Polynomial speedup

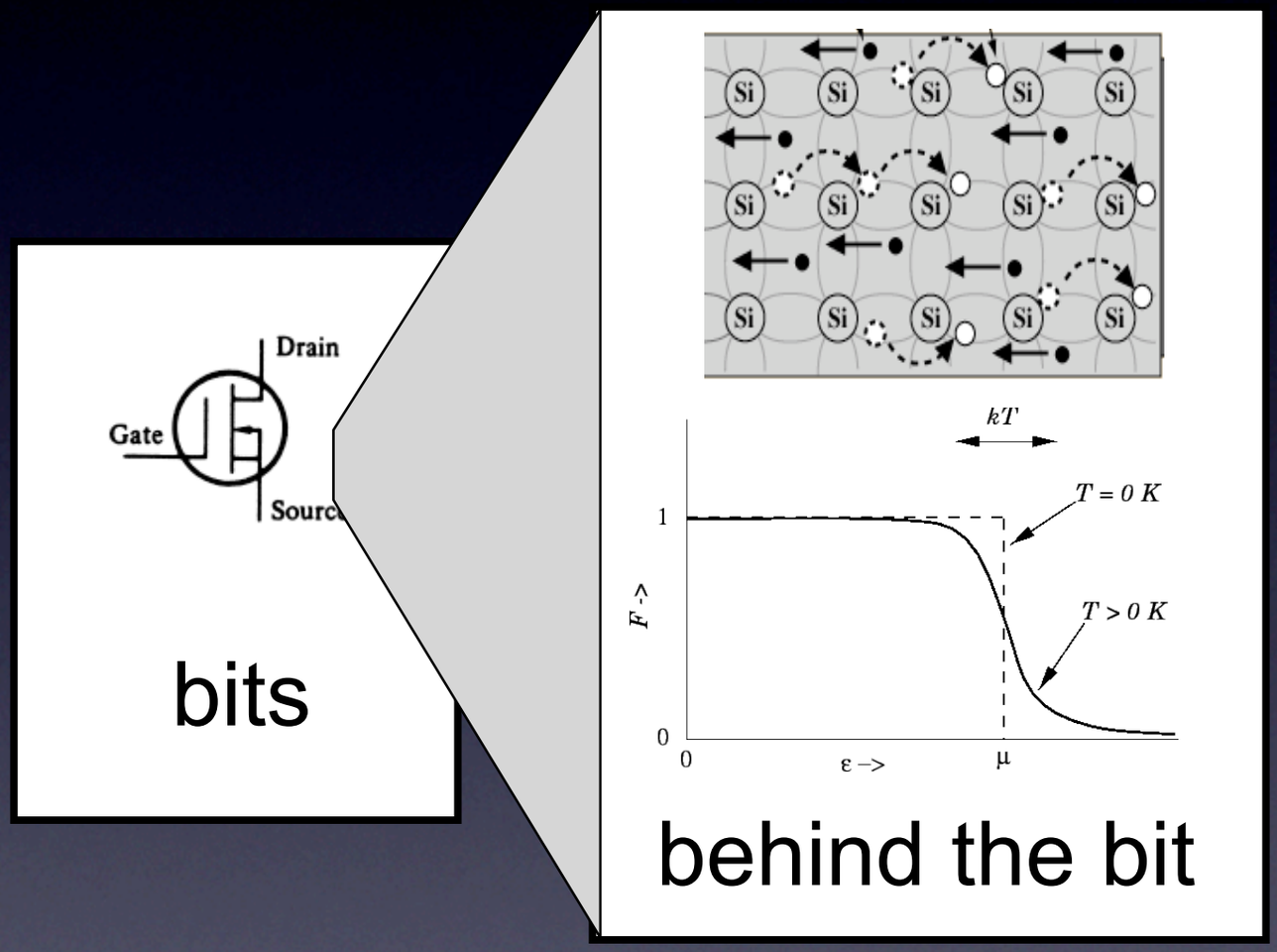
Grover's algorithm
- quadratic speedup in needle-in-haystack search

NAND tree algorithm
- quadratic speedup in game algorithms

Amplitude amplifiction
- Ubiquitous quadratic speedups

A Simple Question

Why is computation physically possible?



Error Correction



b probability 1-p

¬b probability p



000
001
010
100
→0

111
110
101
011
→1

probability fail = $3p^2(1-p)+p^3 < p$ for $p<1/2$

Claude Shannon 1948

Perfection Through Concatenation

failure probability

$$\frac{}{p}$$

Perfection Through Concatenation



failure probability

p

cp^2

Perfection Through Concatenation



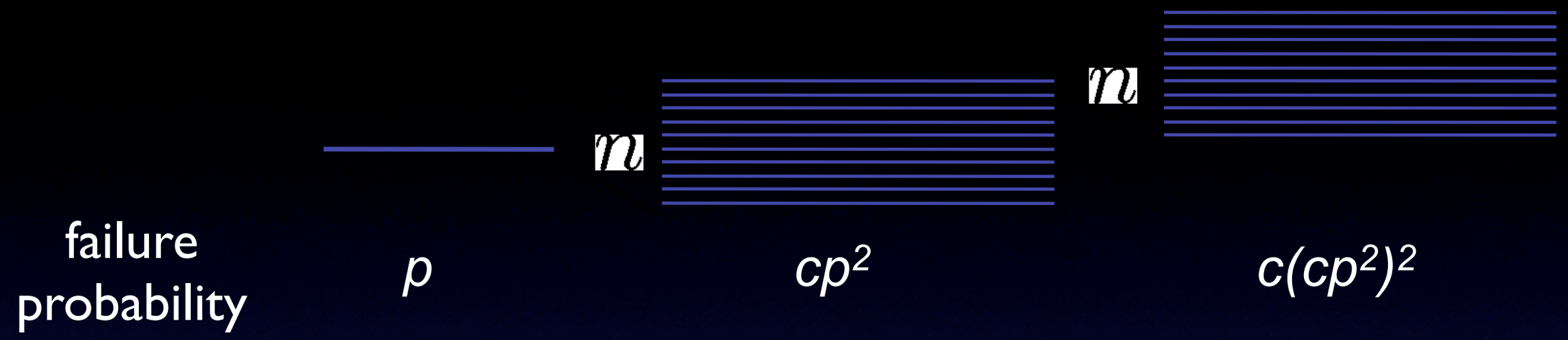
failure probability

p

cp^2

$c(cp^2)^2$

Perfection Through Concatenation



At k^{th} level of concatenation, probability of an error which destroys the encoded information is

$$p_{fail} = \frac{(cp)^{2^k}}{c}$$

$$\frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k}$$

$$p_{th} = \frac{1}{c}$$

and uses $N_{qubits} = n^k$ qubits

Perfection Through Concatenation



At k^{th} level of concatenation, probability of an error which destroys the encoded information is

$$p_{fail} = \frac{(cp)^{2^k}}{c}$$

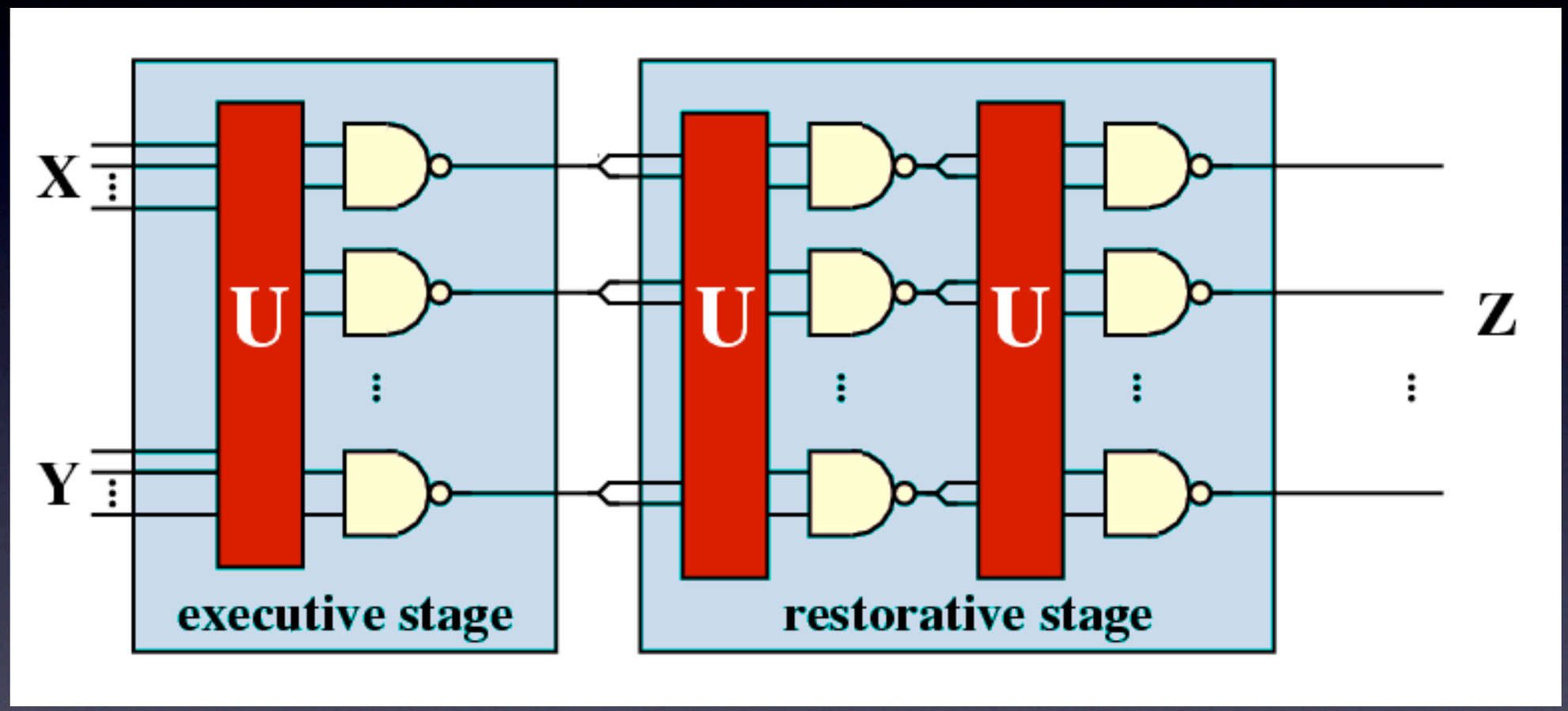
$$\frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k}$$

$$p_{th} = \frac{1}{c}$$

and uses $N_{qubits} = n^k$ qubits

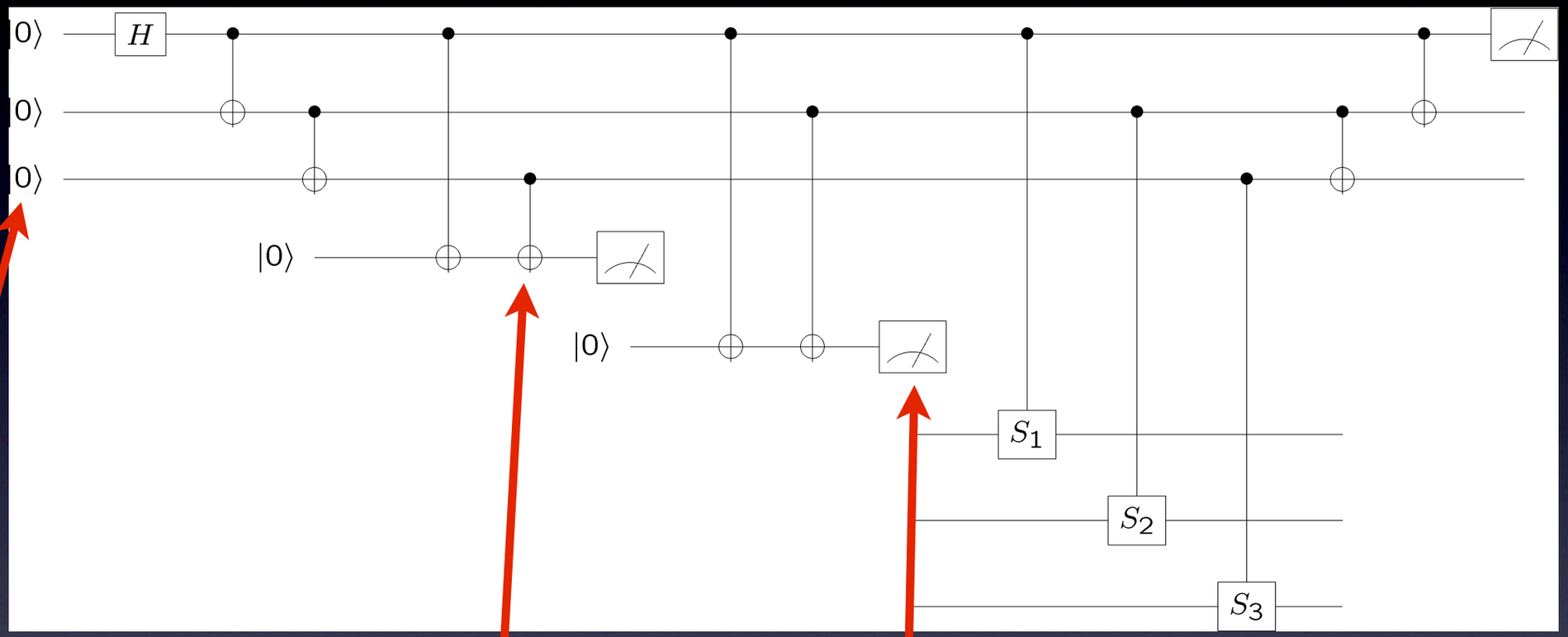
$$N_{qubits} = \left[\log_{\frac{p}{p_{th}}} \left(\frac{p_{th}}{p_{fail}}\right)\right]^{\log_2 n} = O\left(\text{poly}\log \frac{1}{p_{fail}}\right)$$

Fault-Tolerant Computation



von Neumann 1952

Quantum Circuit



Preparation

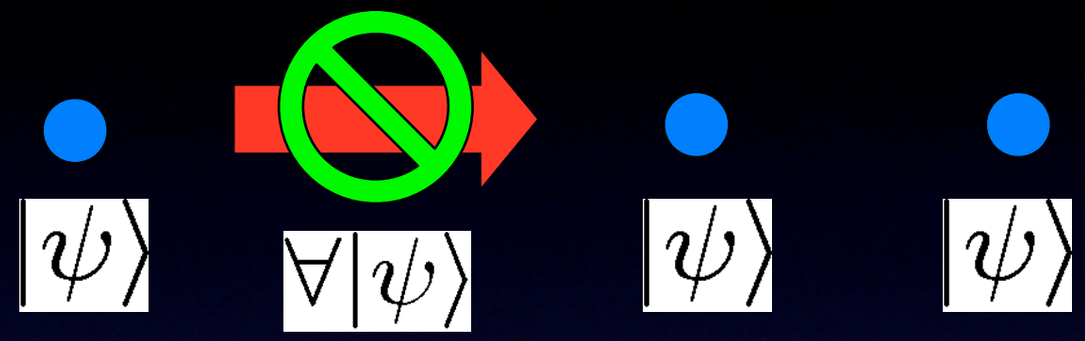
Gates

Measurement

Time

Quantum Challenges

Quantum info
cannot be copied:



"no fan-out"

Quantum info
likes to become
Classical:



010001110101..



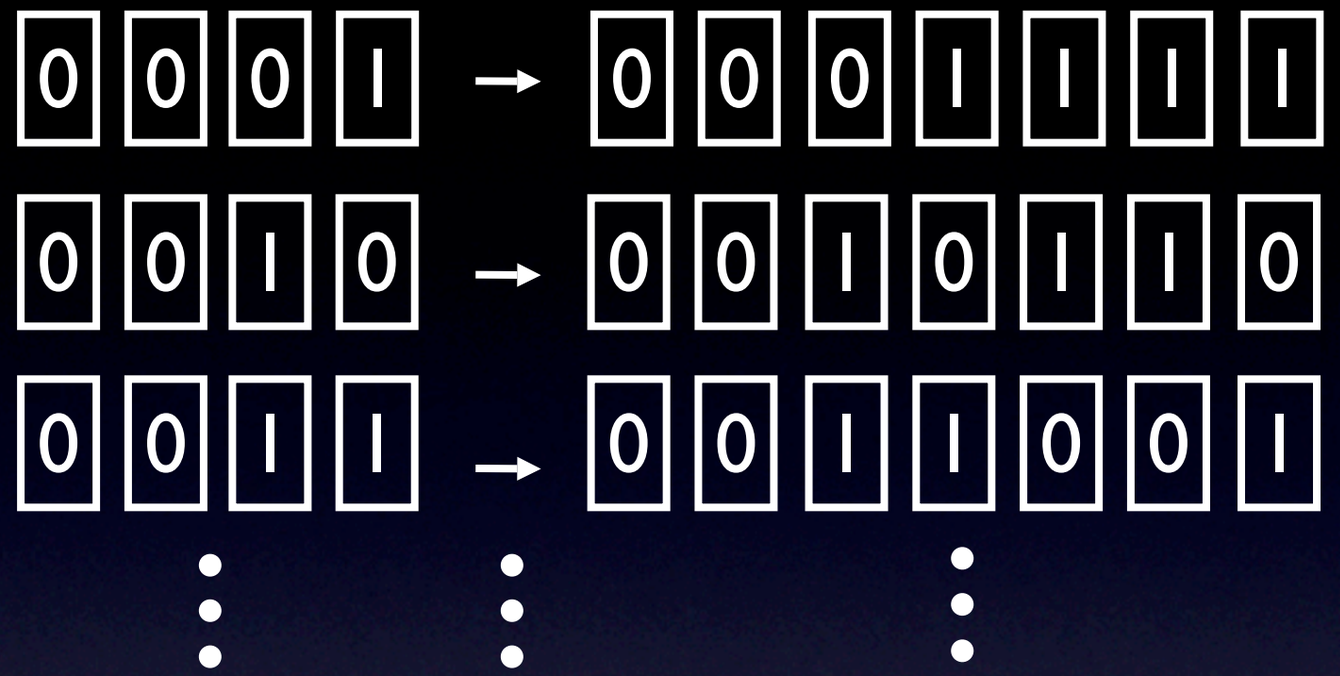
010011110001..

"noisy
channel"

Error Correction

classical error correcting code:

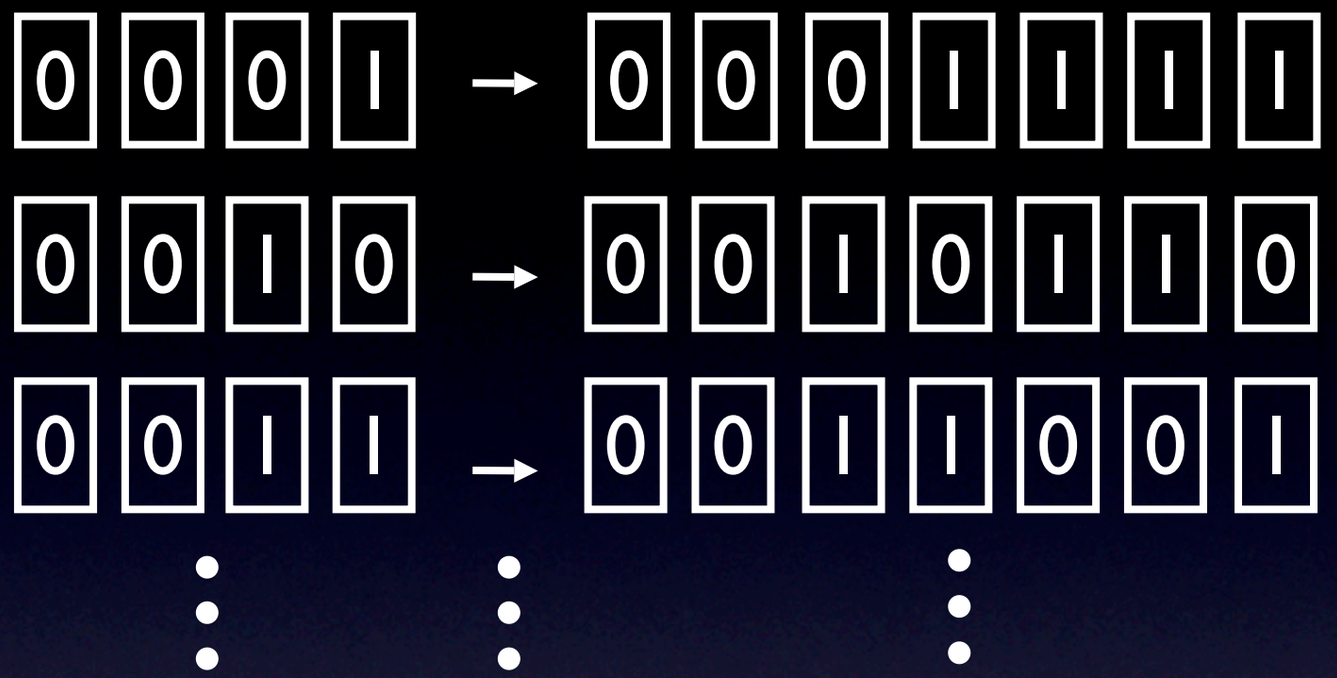
(bits and pbits)

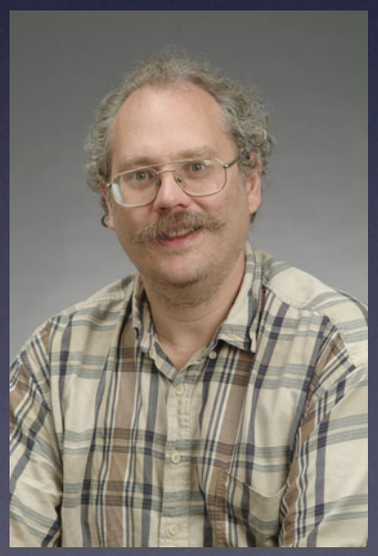


Error Correction

classical error correcting code:

(bits and pbits)





Peter Shor



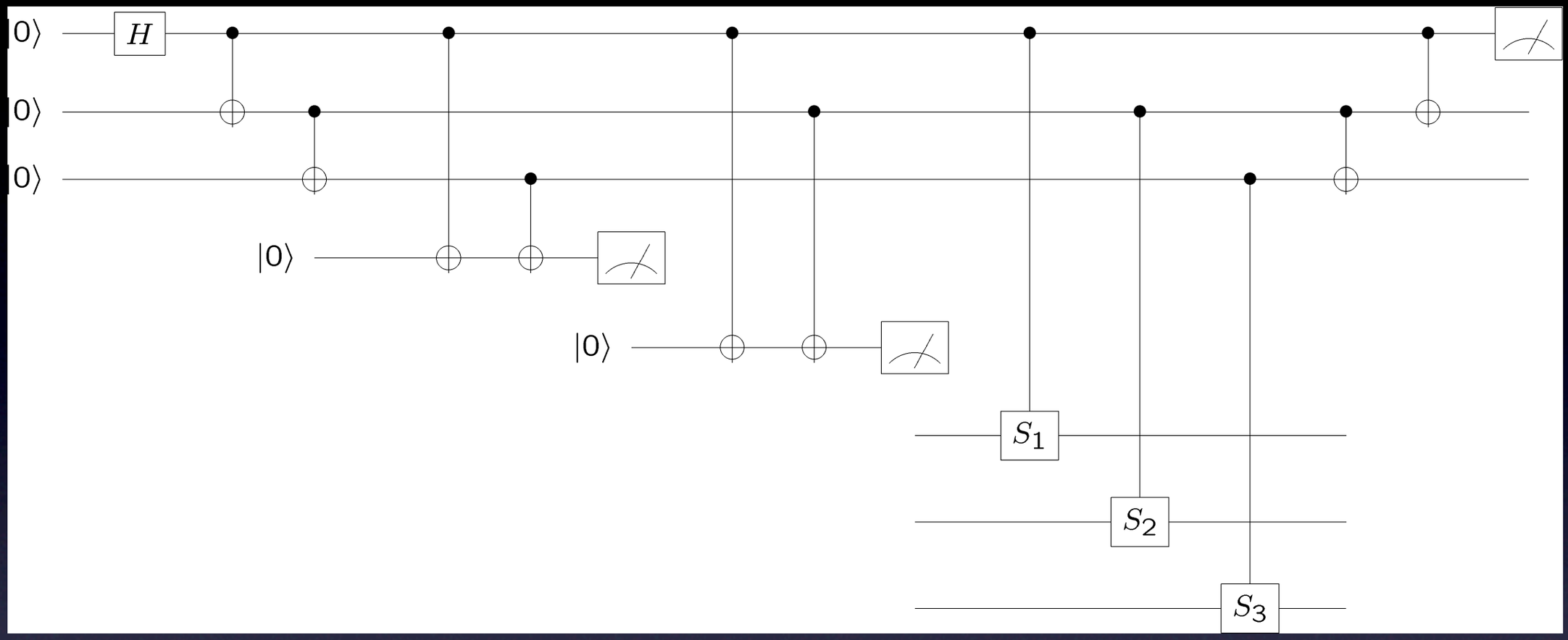
Andrew Steane

1995: quantum error correcting codes

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \rightarrow \square\square\square\square\square$$

encode quantum information across multiple quantum subsystems

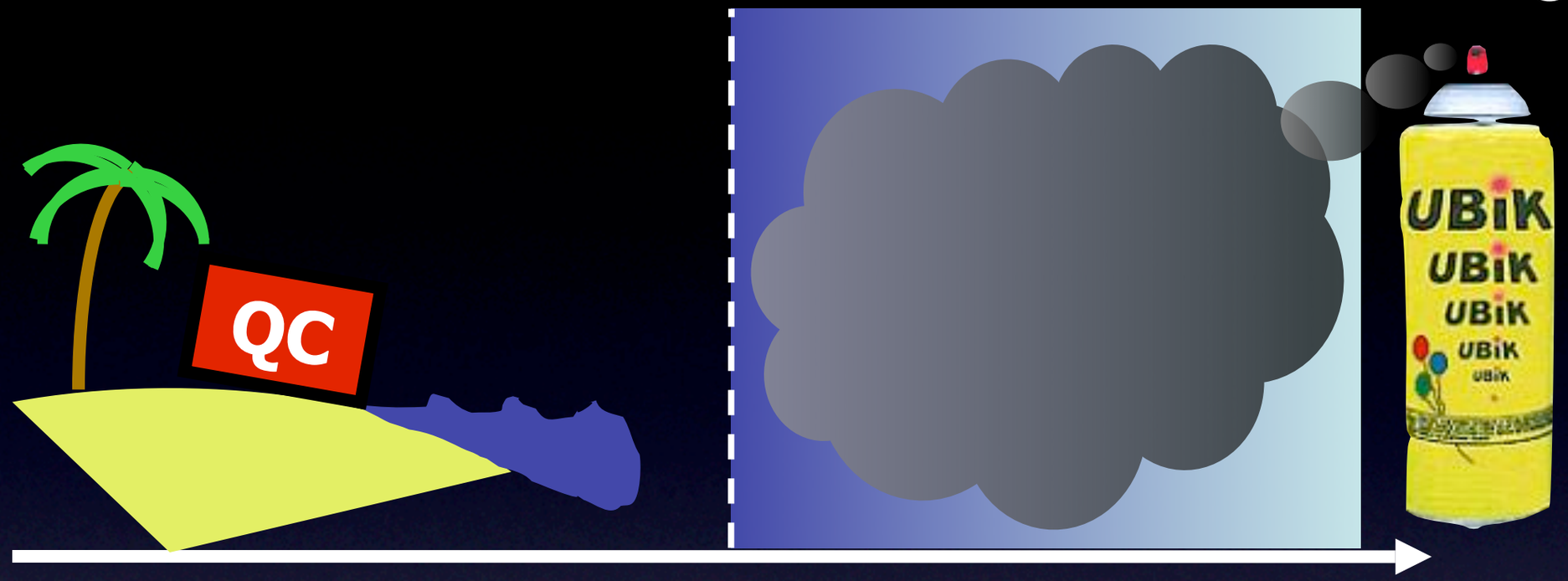
Fault-Tolerant Quantum Computation



Must design quantum circuits which succeed even when every element of the circuit (preparation, unitary gates, Measurements) can fail.

Quantum error correction + Fault-tolerant procedures

Threshold Theorem, 1996



"decoherence"

"imprecision"

Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Gottesman, Preskill]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Architectures For Quantum Computers

Systems-level challenges to achieving scalable, fault-tolerant quantum computation



Error Correction

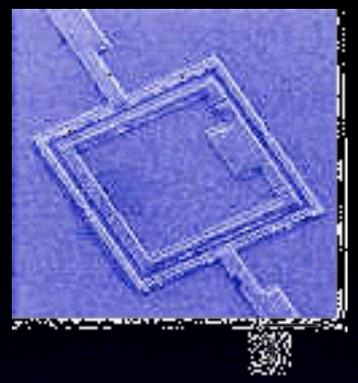
Designing computer architectures when components fail

The error correcting architectures must be massively (but fairly embarrassingly) parallel

Building Blocks

stationary

fixed location

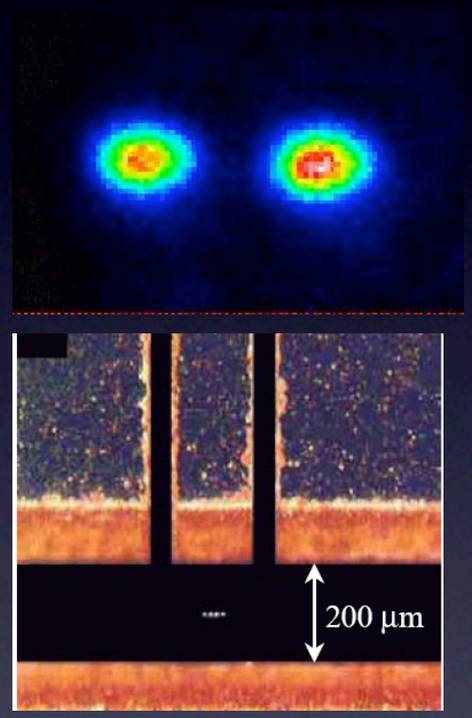


Example:
superconducting qubits

qubits

walking

"slow"



Example:
ion trap qubits

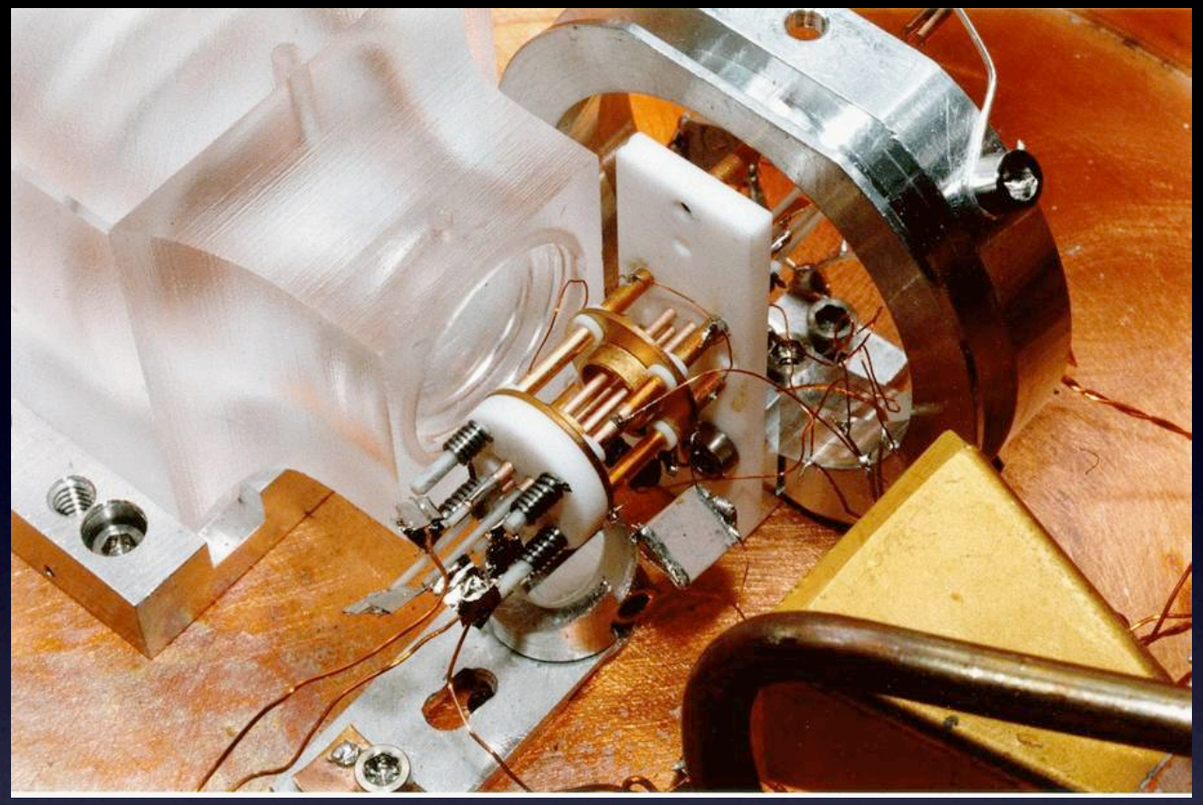
flying

"fast"



Example:
photons

Ion Trap

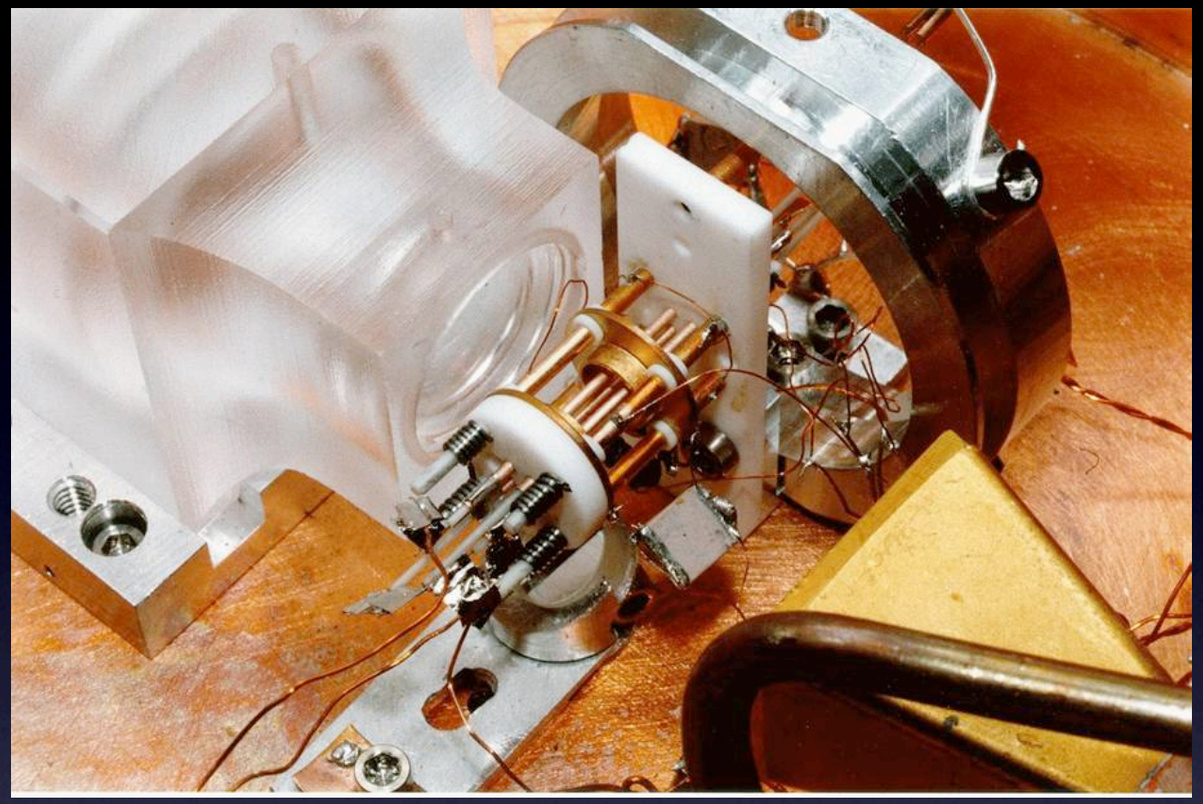


Oscillating electric fields trap ions

like charges repel



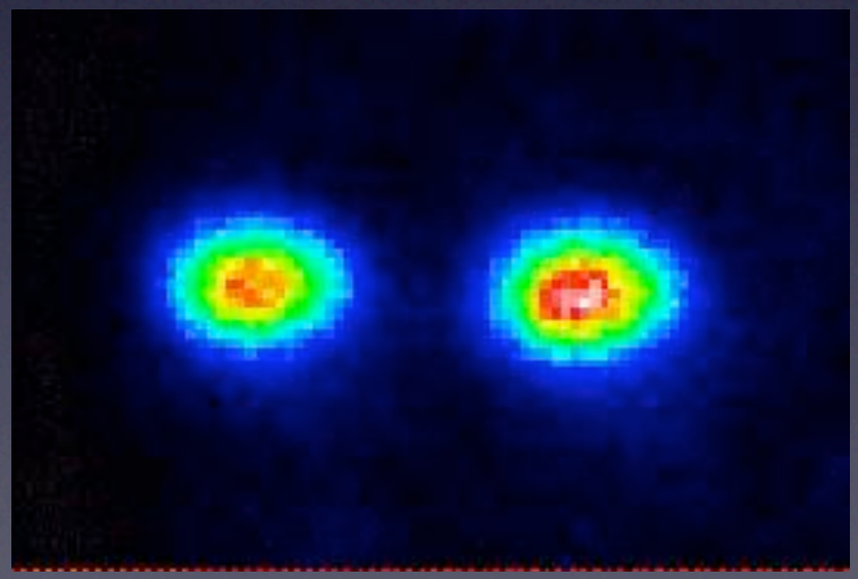
Ion Trap



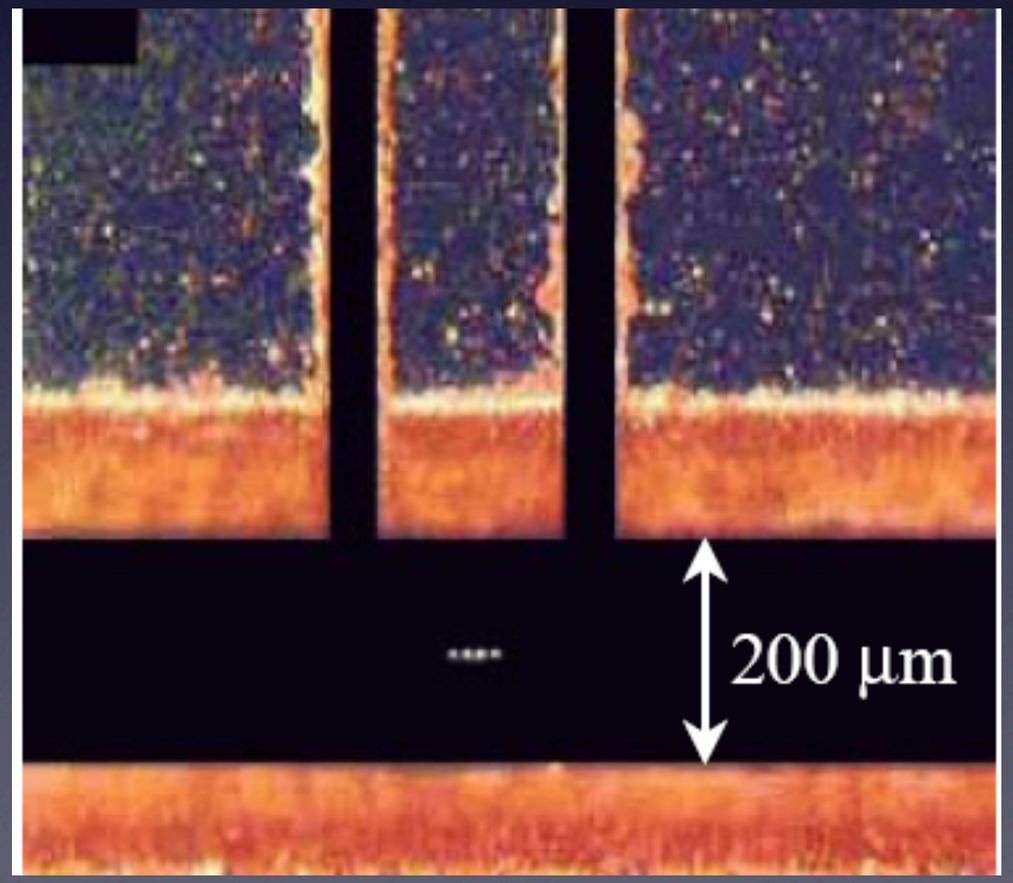
Oscillating electric fields trap ions

like charges repel

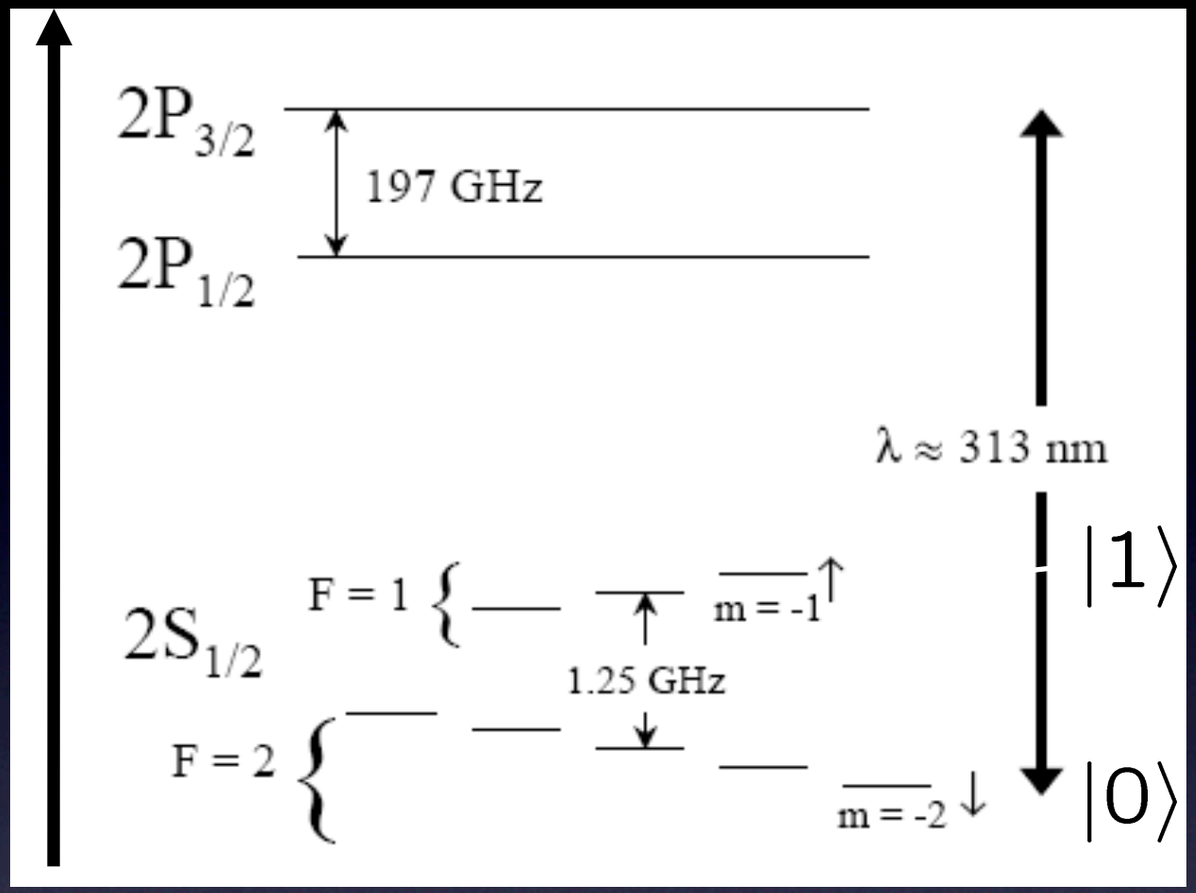


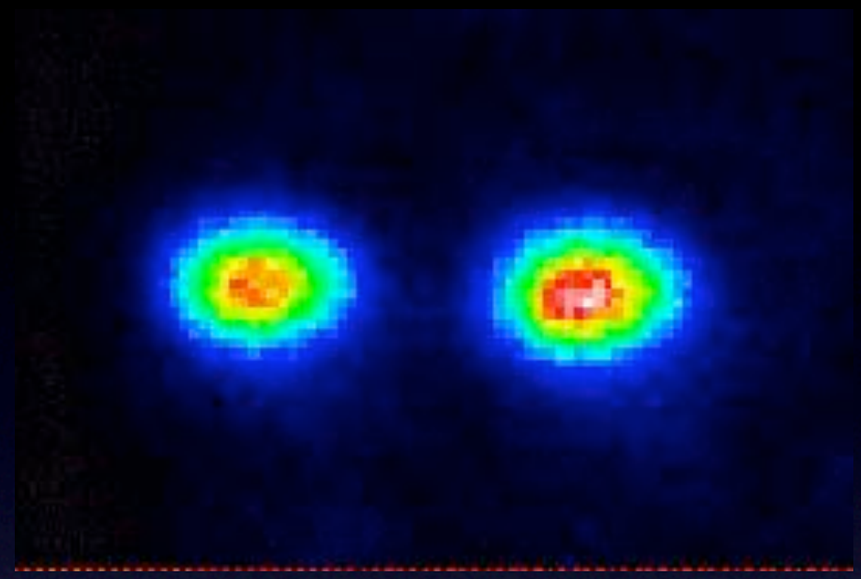


2 $^{9}Be^{+}$ Ions in an Ion Trap



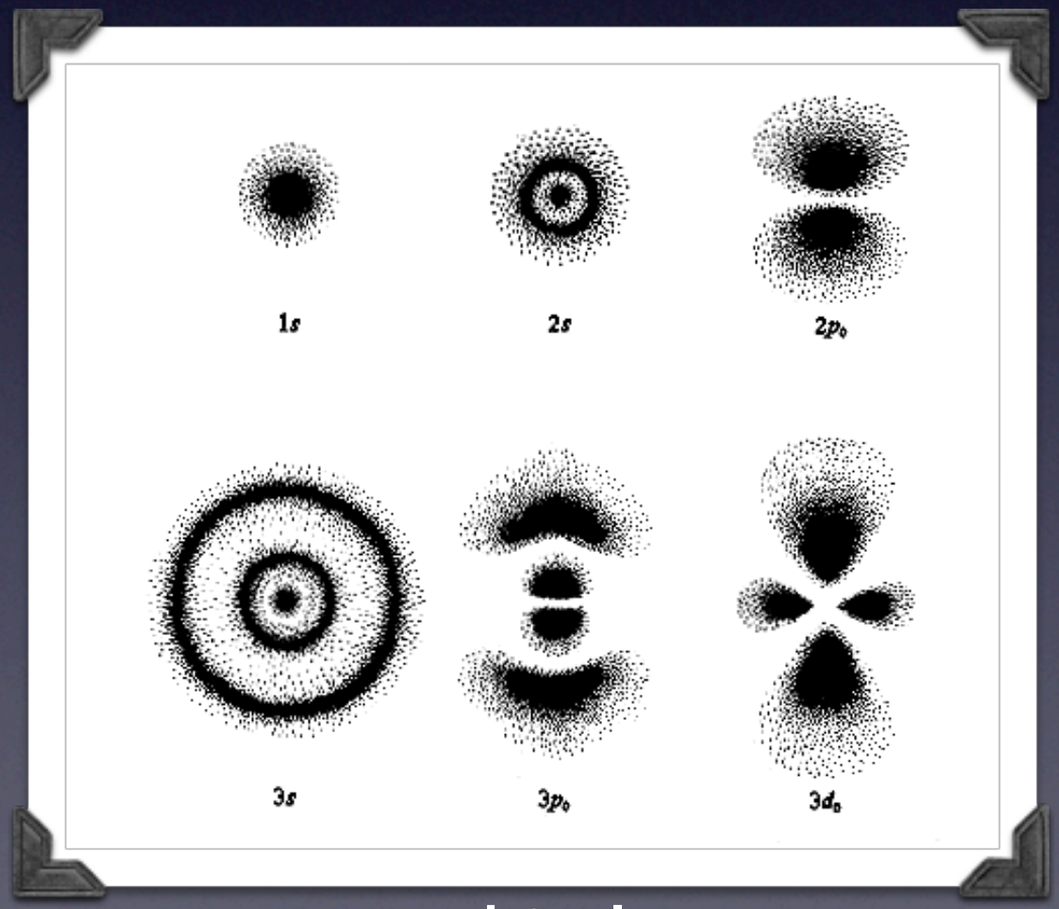
Where's the Qubit?





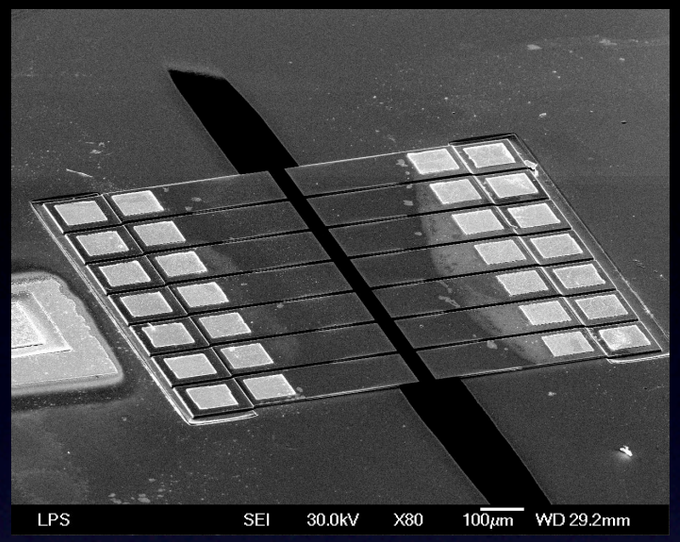
Each ion = 1 qubit

1. Well defined qubits

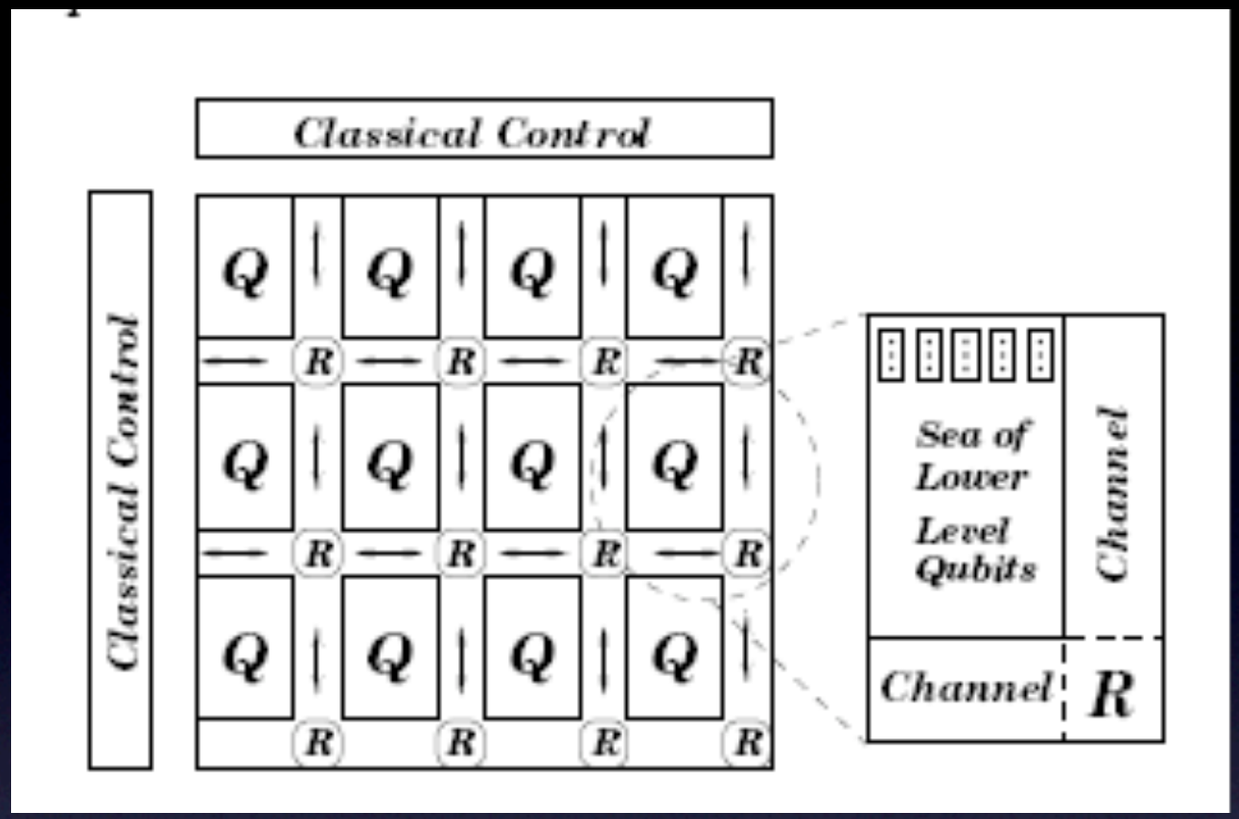


orbitals

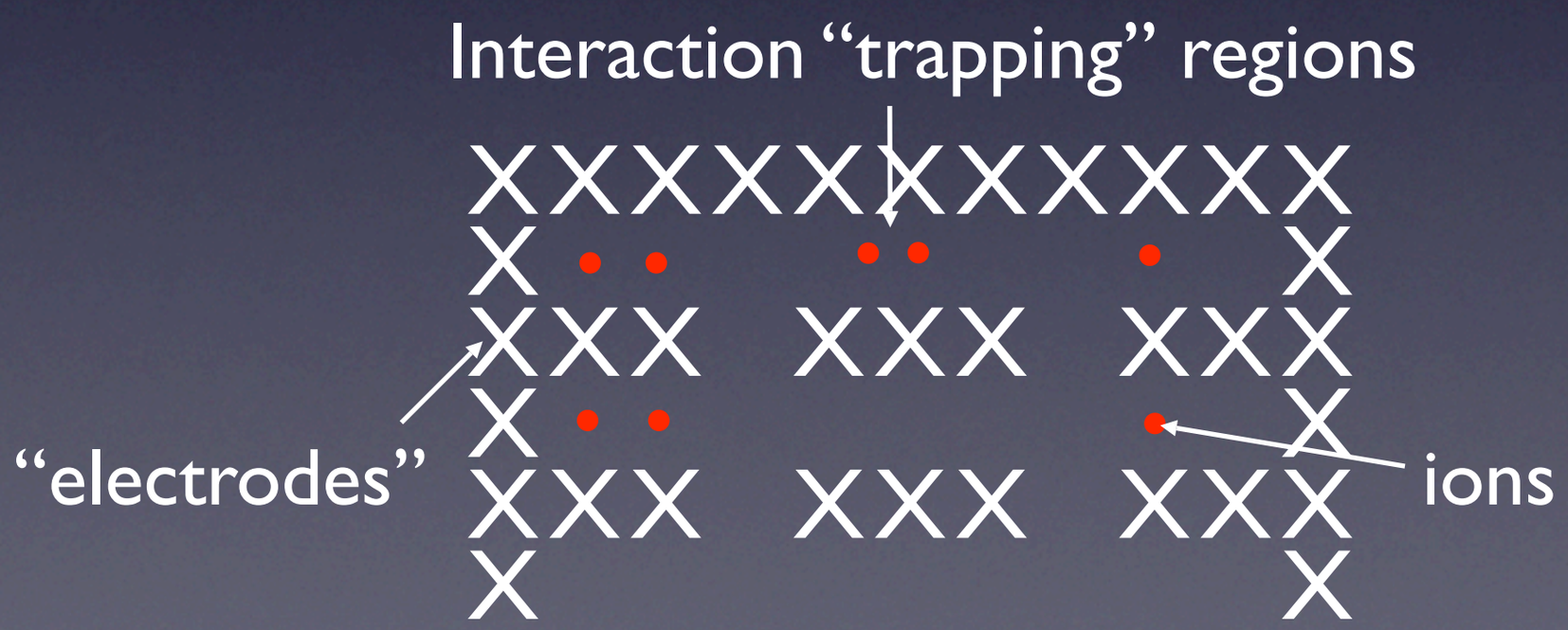
Sketch Ion Trap QC



http://iontrap.physics.lsa.umich.edu/research/trap



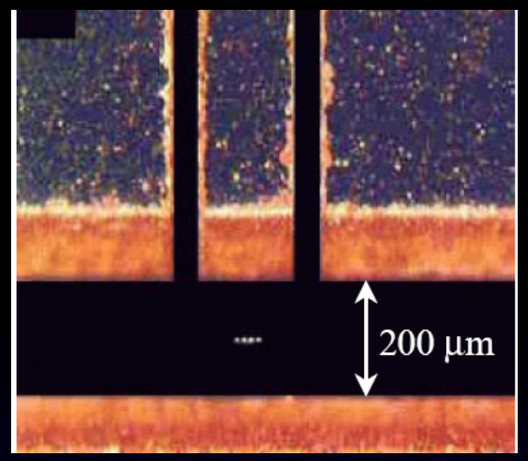
(from Metodi et al, MICRO-38)



Time Scales

walking

"slow"



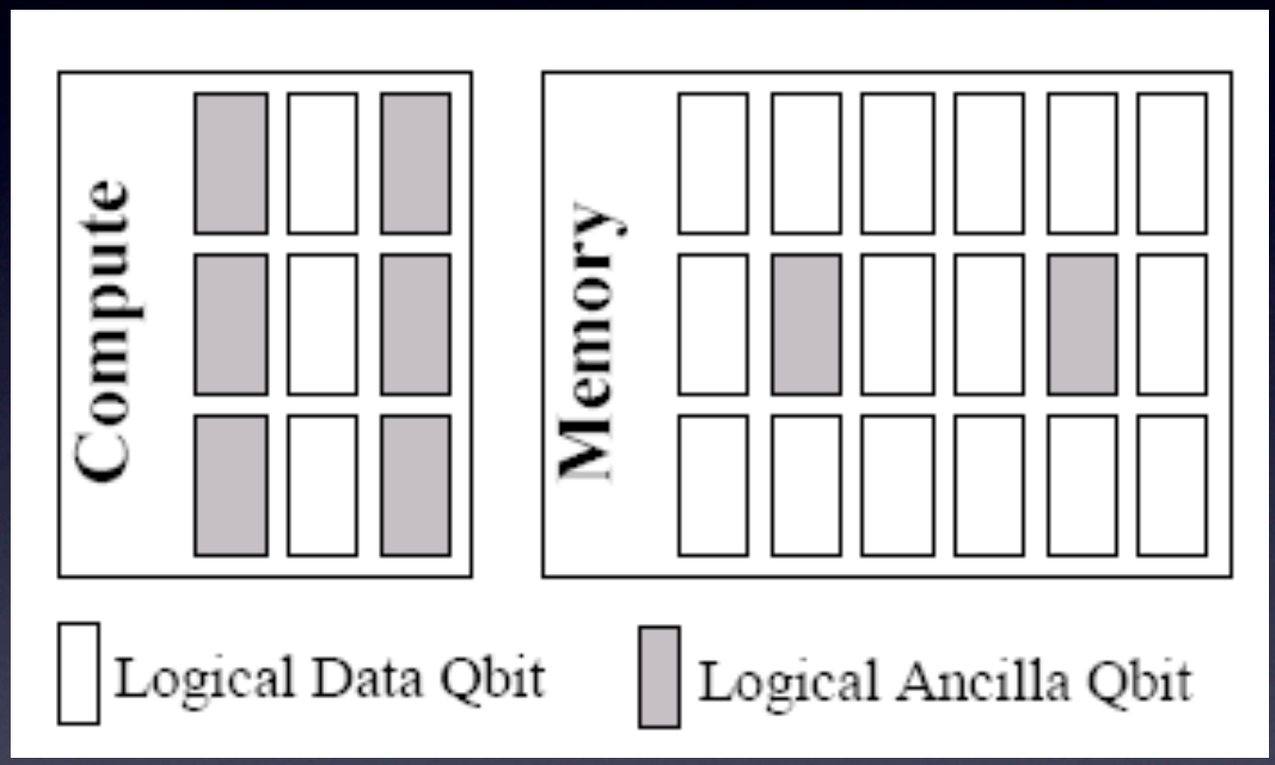
Example:
ion trap qubits

Operation	Time (μs)	Failure rate (today)	Failure rate (future)
single qubit gate	1	10^{-4}	10^{-8}
two qubit gate	10	10^{-2}	10^{-7}
measurement	200	10^{-2}	10^{-8}
movement	20	10^{-3}/ μm	10^{-7}/ μm
cooling	200	?	?
wait	10^{7} / error		

Quantum Memory Hierarchy

Memory error rates << Errors during computation

Error correction more frequent during computation

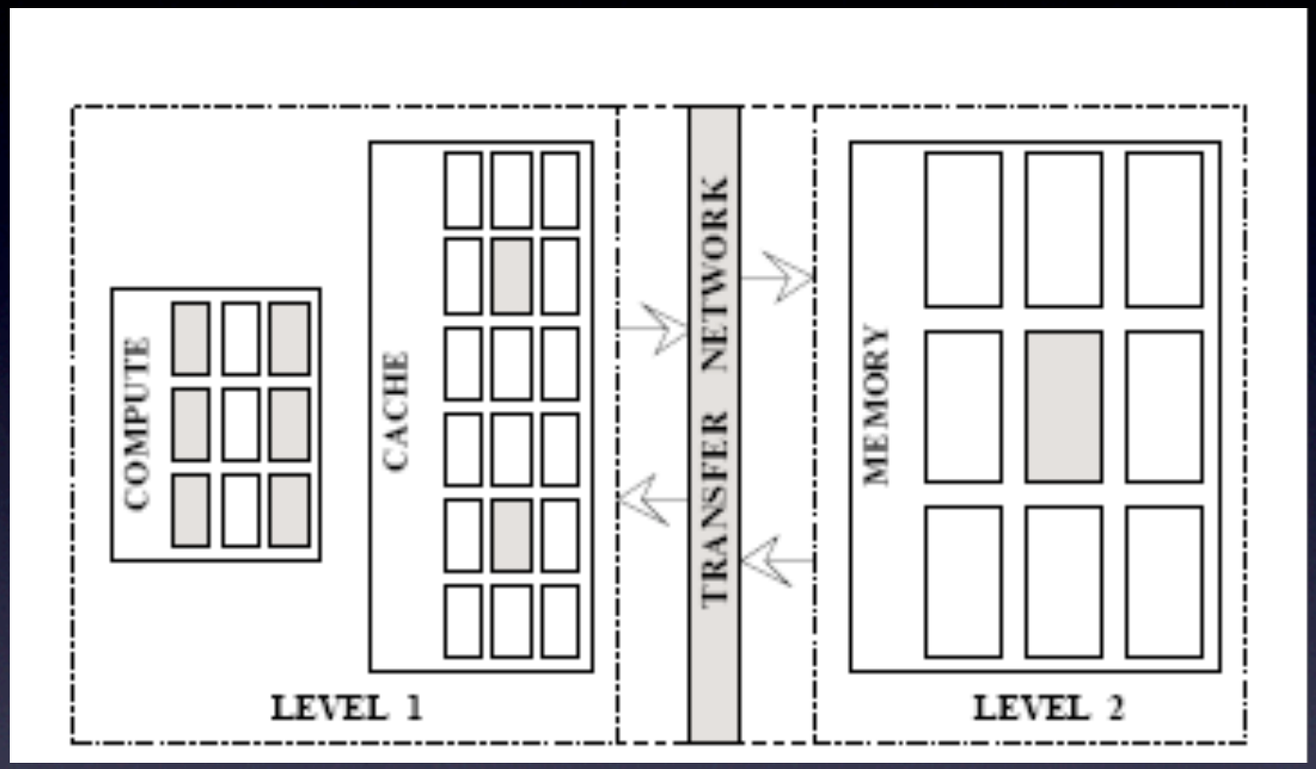


More ancillas (used for error correction) in compute region

Tradeoffs versus Size (Area)

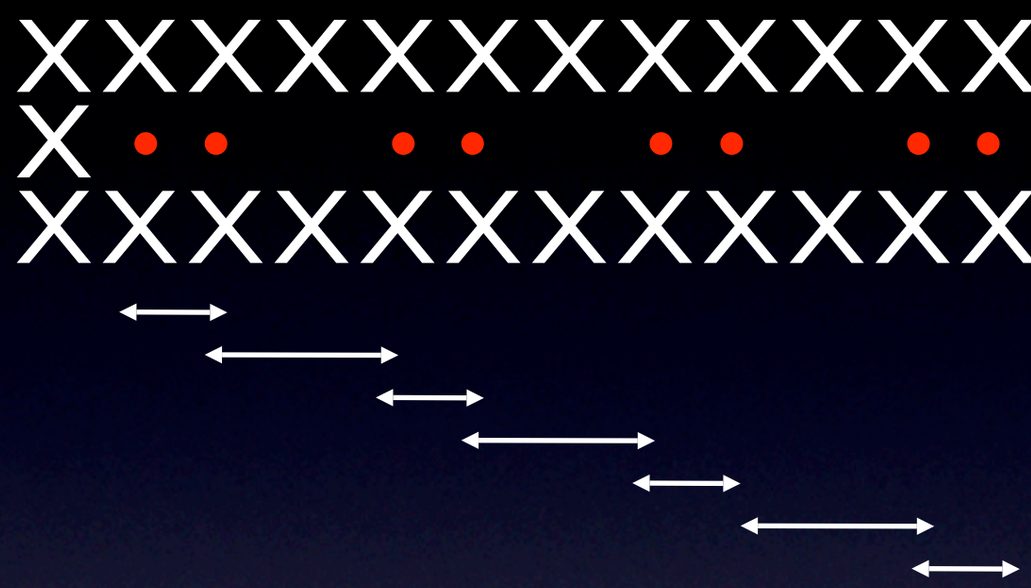
Quantum Memory Hierarchy

Slow memory / Fast compute leads to standard cache ideas…



except…

Communication



$P_f=(p_0)^k$

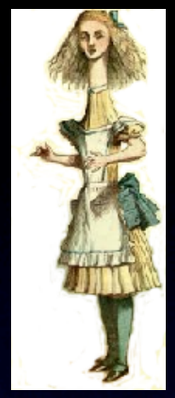
Exponential decrease in distance

Distance k

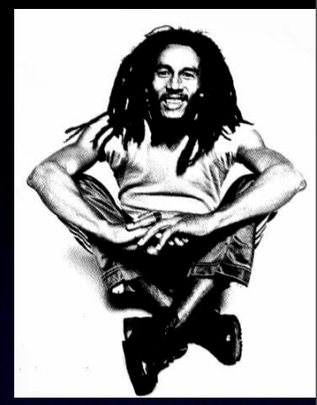
Could perform error correction every k_0 steps….slow!

Quantum Teleportation

Alice wants to send $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ to Bob

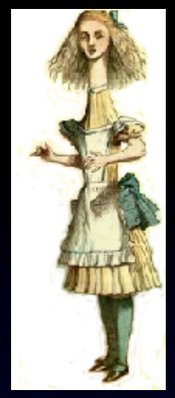


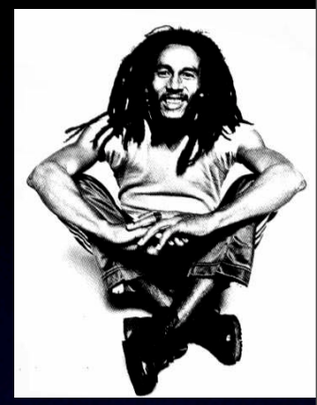
$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$



Quantum Teleportation

Alice wants to send $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ to Bob





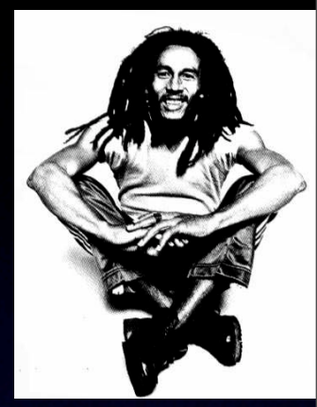
$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

1 qubit

Quantum Teleportation

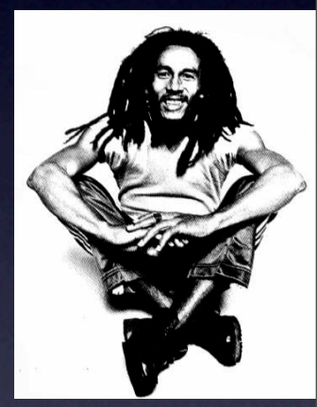
Alice wants to send $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ to Bob

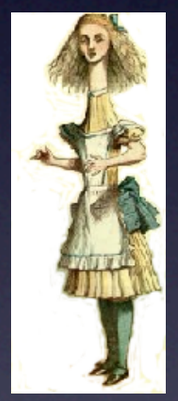




$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

1 qubit





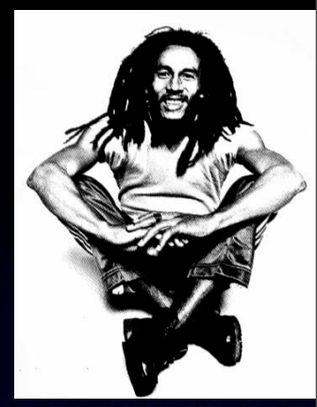
$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

$$|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Quantum Teleportation

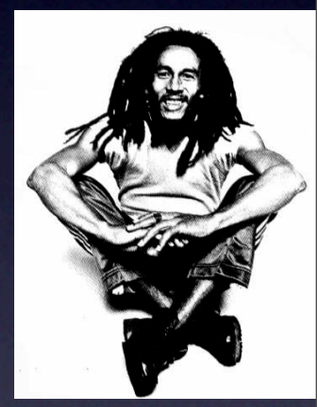
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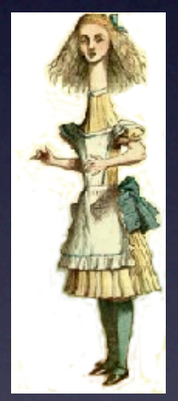




$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

1 qubit





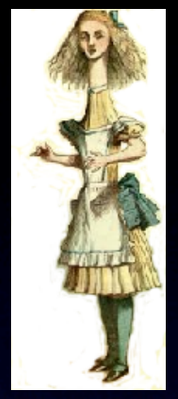
$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

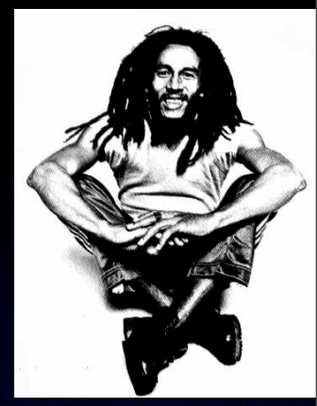
$$|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Alice performs operation on her state and half of entangled system and sends outcome

Quantum Teleportation

Alice wants to send $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ to Bob

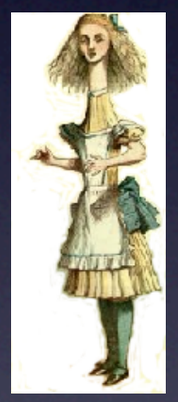




$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

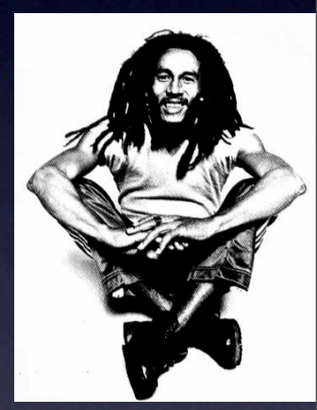
1 qubit

2 bits



$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

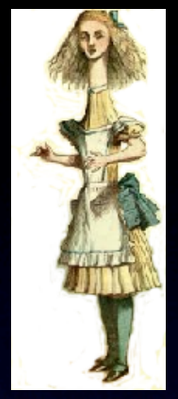
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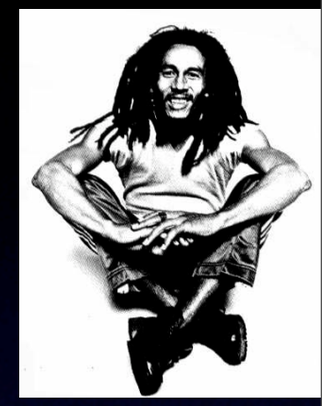


Alice performs operation on her state and half of entangled system and sends outcome

Quantum Teleportation

Alice wants to send $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ to Bob

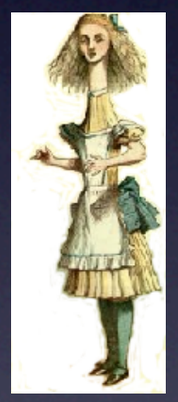




$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

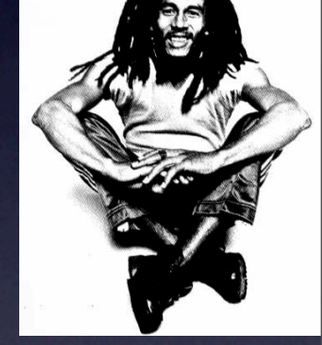
1 qubit

2 bits



$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

$$|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$



$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

Alice performs operation on her state and half of entangled system and sends outcome

Conditional on received bits, Bob performs manipulation of his half of entangled system

Moving Qubits

Teleportation channel

classical control



entangled qubits

Entanglement distillation



classical control

Moving Qubits

Teleportation channel

classical control



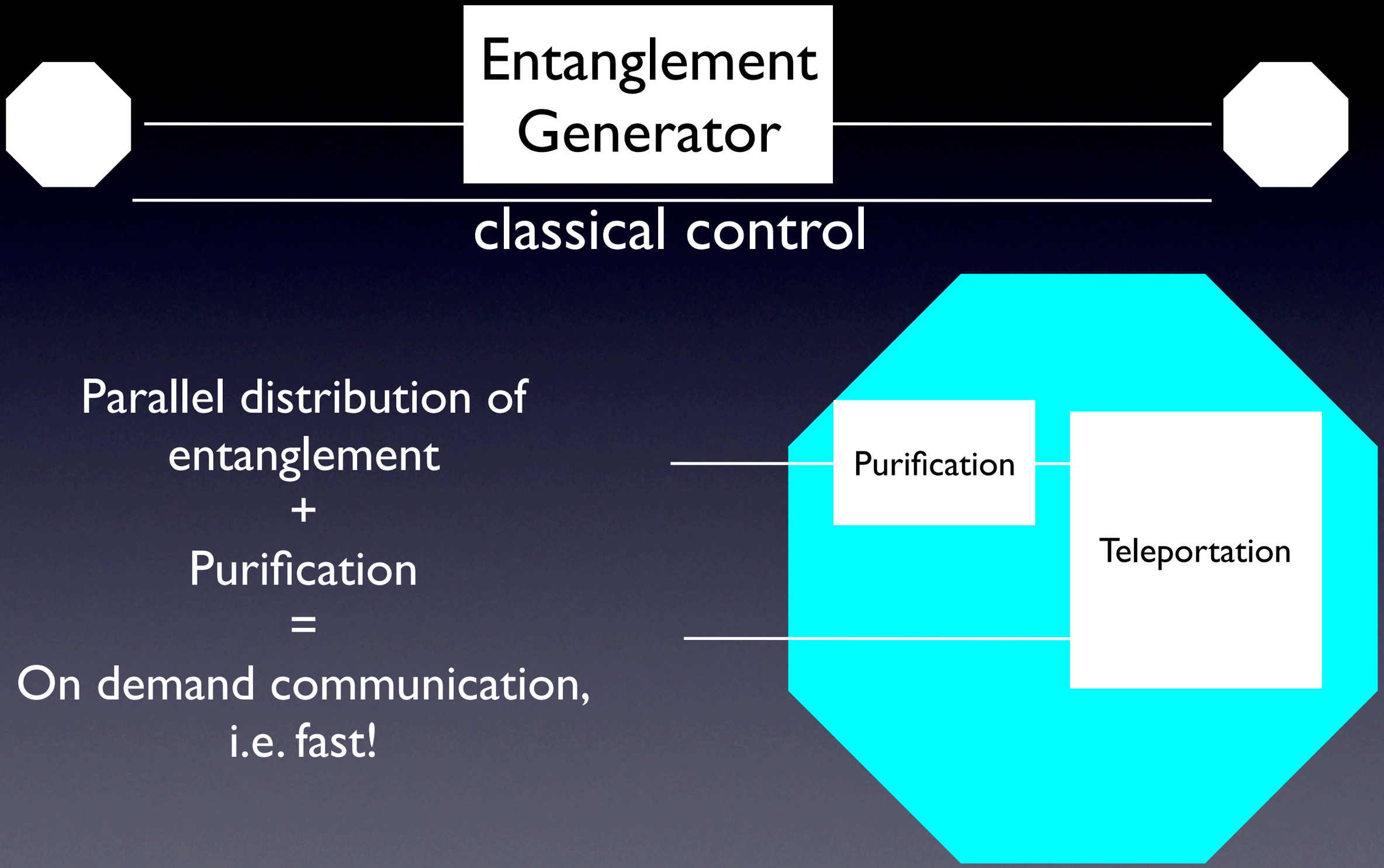
entangled qubits

Entanglement distillation



classical control

Teleportation Units

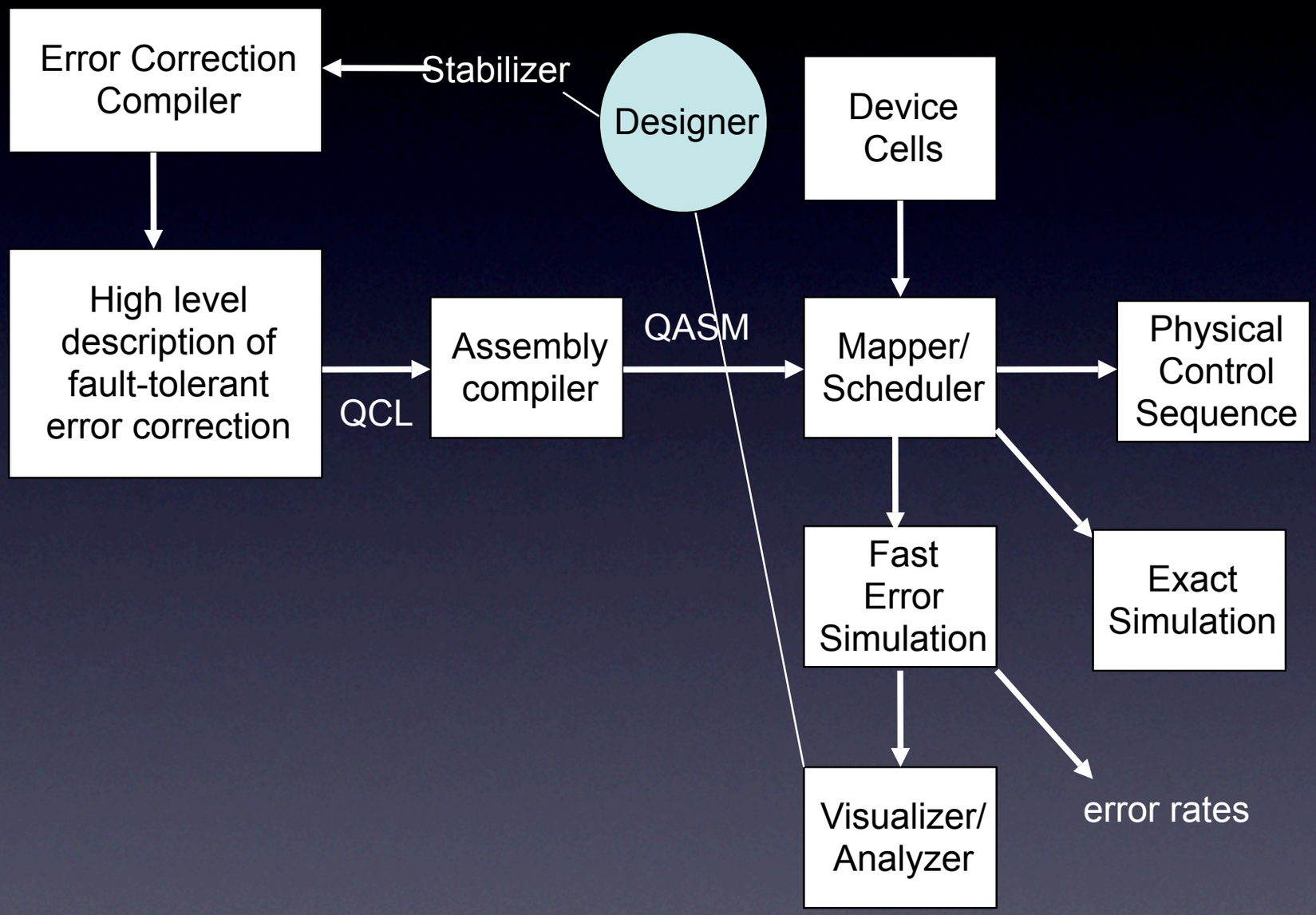


Parallel distribution of entanglement
+
Purification
=
On demand communication, i.e. fast!

Quantum Architectures

New generation of tools in order to aid in the design of quantum computing architectures….

Tools!


Error Correction Compiler
Stabilizer
Designer
Device Cells
High level description of fault-tolerant error correction
QCL
Assembly compiler
QASM
Mapper/ Scheduler
Physical Control Sequence
Fast Error Simulation
Exact Simulation
Visualizer/ Analyzer
error rates

High Level Languages

Example in QCL (http://tph.tuwien.ac.at/~oemer/qcl.html)

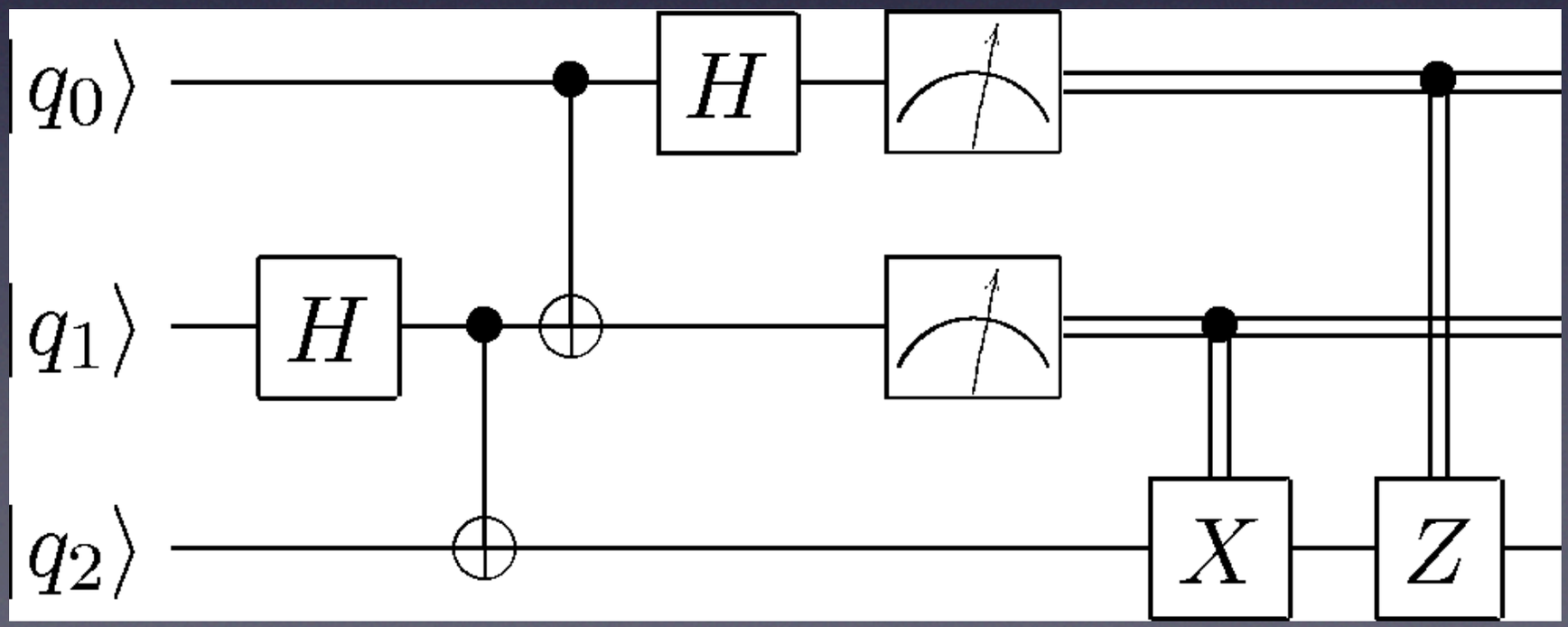
```
operator dft(qureg q) {  // main operator
const n=#q;              // set n to length of input
int i; int j;            // declare loop counters
for i=0 to n-1 {
   for j=0 to i-1 {      // apply conditional phase gates
      CPhase(2*pi/2^(i-j+1),q[n-i-1] & q[n-j-1]);
   }
   Mix(q[n-i-1]);        // qubit rotation
}
flip(q);                 // swap bit order of the output
}
```

Quantum Assembly Language

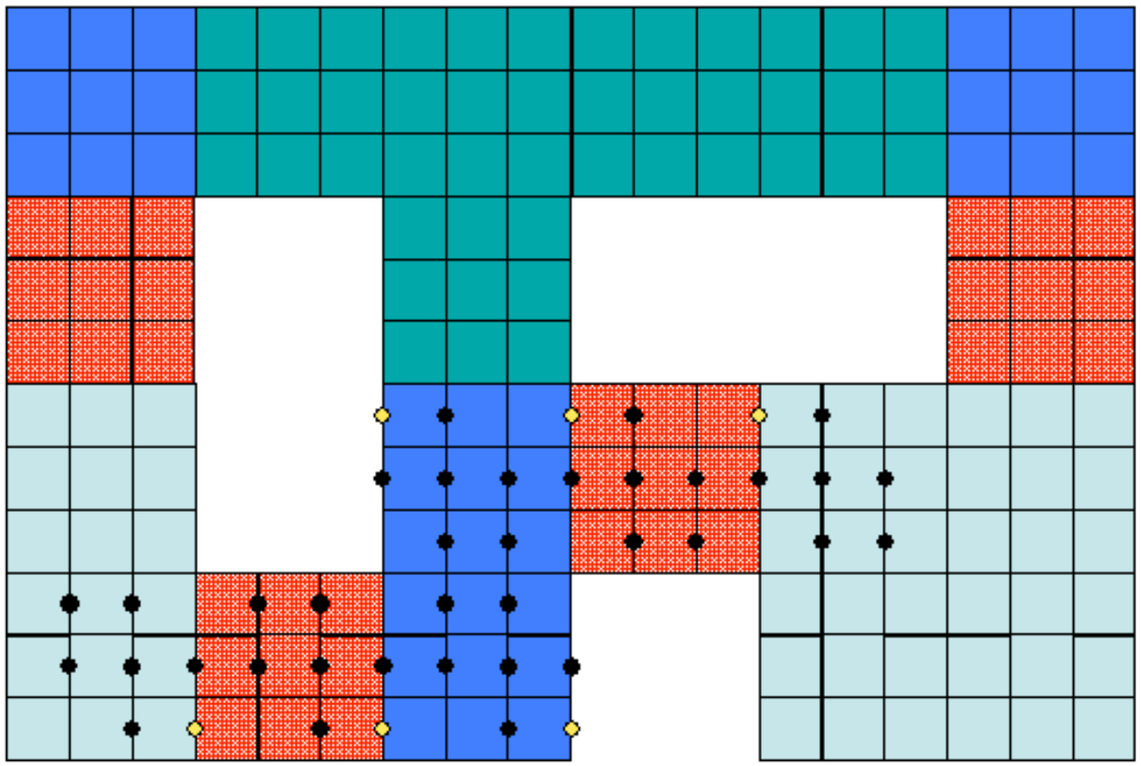
```
qubit q0
qubit q1
qubit q2

h q1            # create EPR pair
cnot q1,q2
cnot q0,q1      # Bell basis measurement
h q0
nop q1
measure q0
measure q1
c-x q1,q2       # correction step
c-z q0,q2
```

QASM



Scheduler



Z/X 1-EC

Example from Chuang group at http://www.media.mit.edu/quanta

Further Resources

Quantum Computing For Computer Scientists (David Mermin):
http://people.ccmr.cornell.edu/~mermin/qcomp/CS483.html

"Standard" Textbook:
Quantum Computation and Quantum Information
by Michael Nielsen and Isaac Chuang

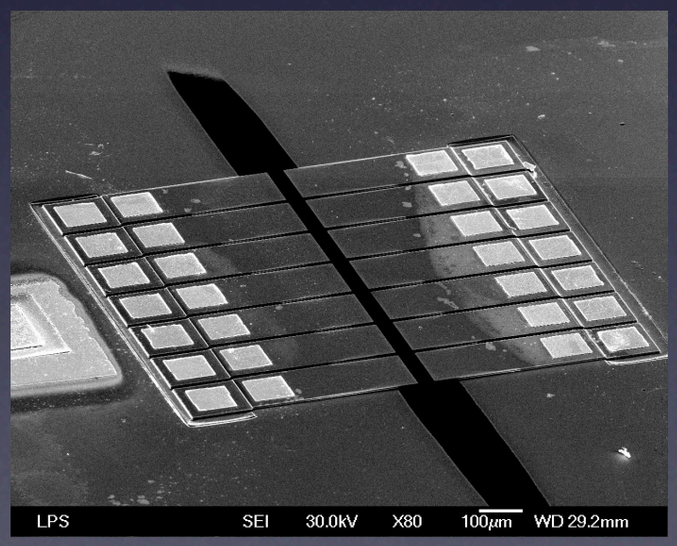
ISCA 2002 Tutorial Page (Oskin and Chong):
http://www.cs.washington.edu/homes/oskin/quantum-tutorial/

Quantum Architectures



Today: a taste

....becoming extremely important



5-8 qubit ion trap machines

microfabricated ion traps

large effort to scale up to traps with 30-100 in next few year

Picture from http://iontrap.physics.lsa.umich.edu/research/trap

DiVincenzo's Criteria



David DiVincenzo

1. Well defined qubits in a scalable architecture

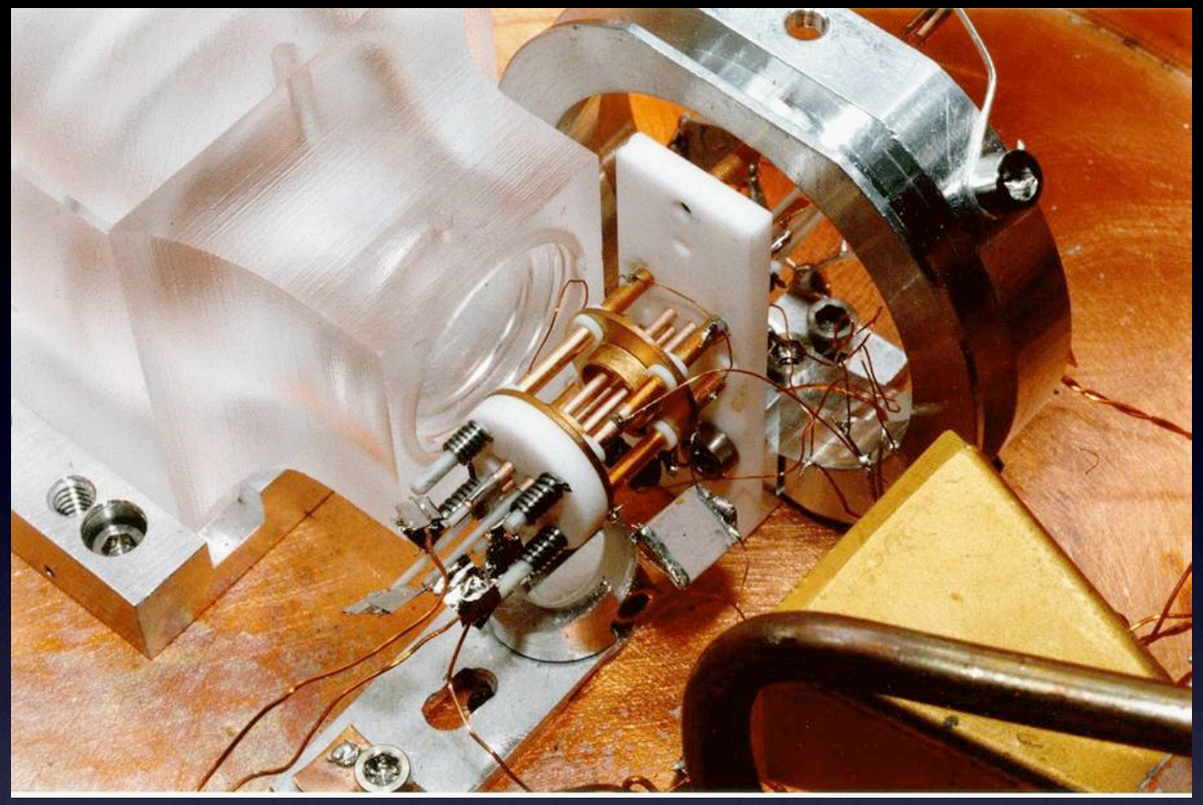
2. The ability to initialize the system to a fixed wave function.

3. Have faster control over the system than error processes in the system.

4. Have the ability to perform a universal set of quantum gates.

5. Have the ability to perform high quality measurements

Ion Trap

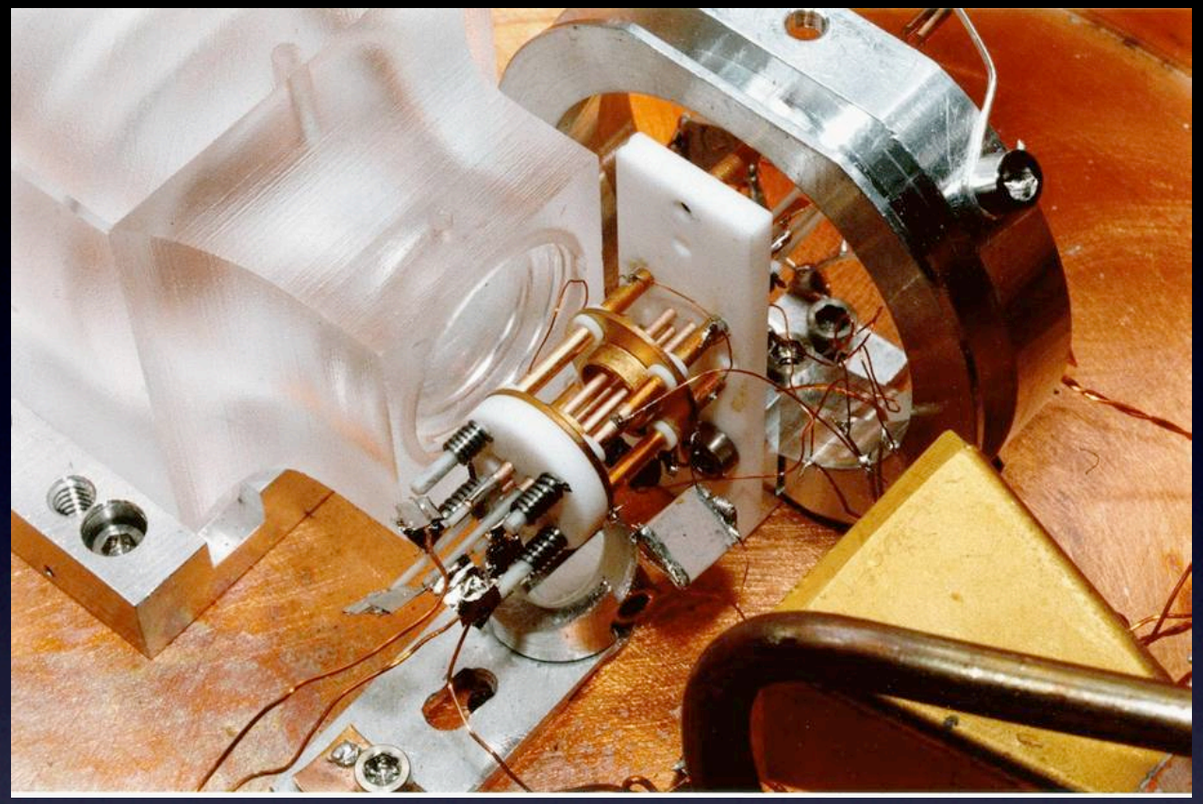


Oscillating electric fields trap ions

like charges repel



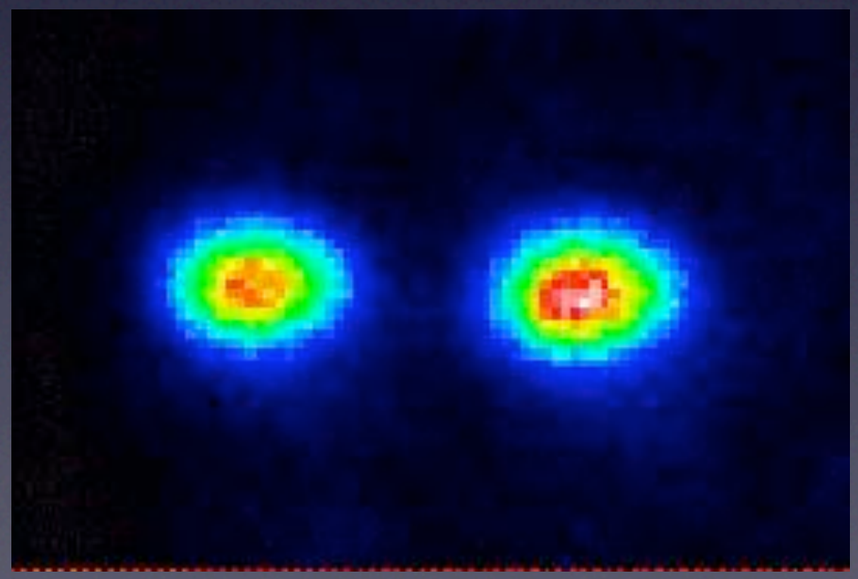
Ion Trap



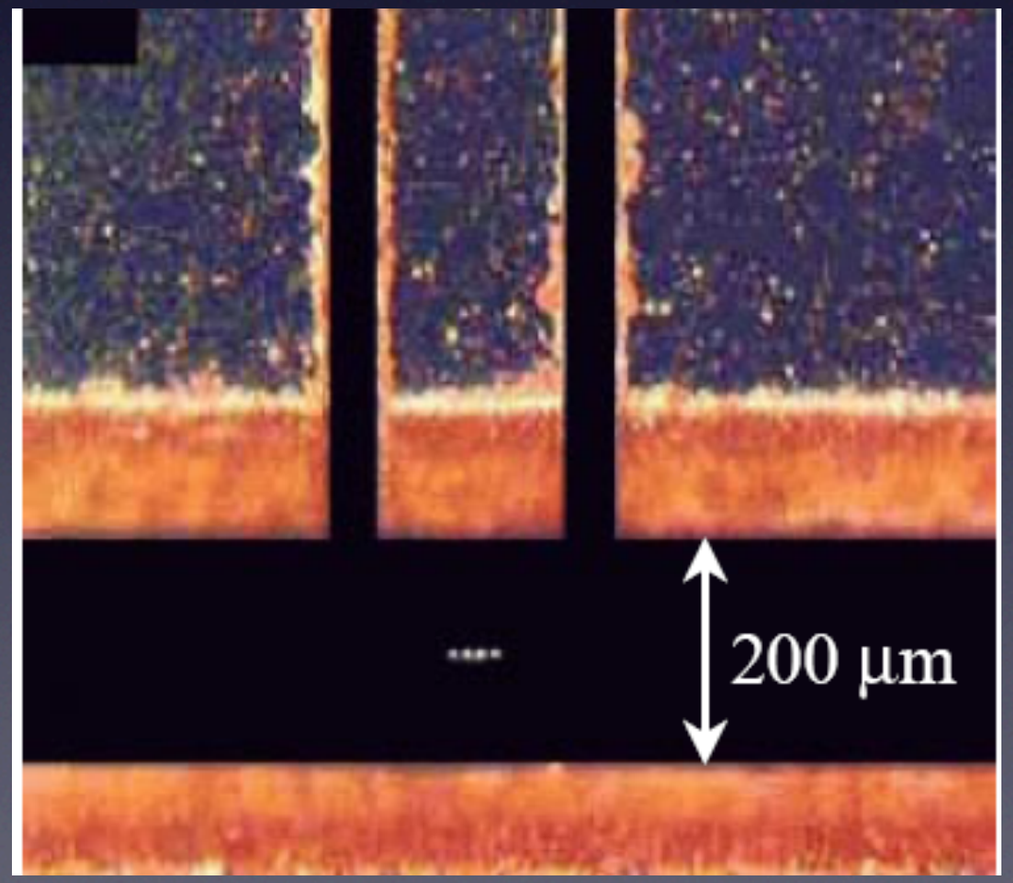
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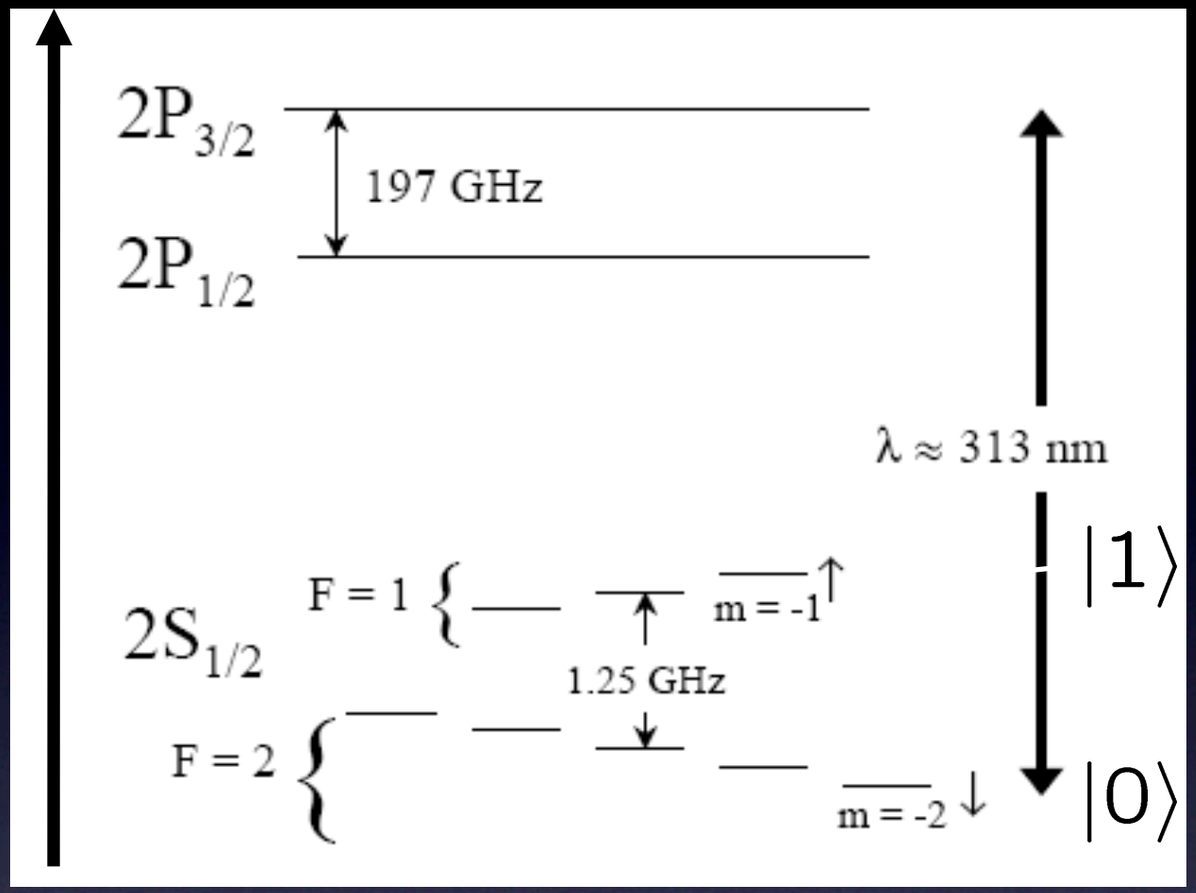


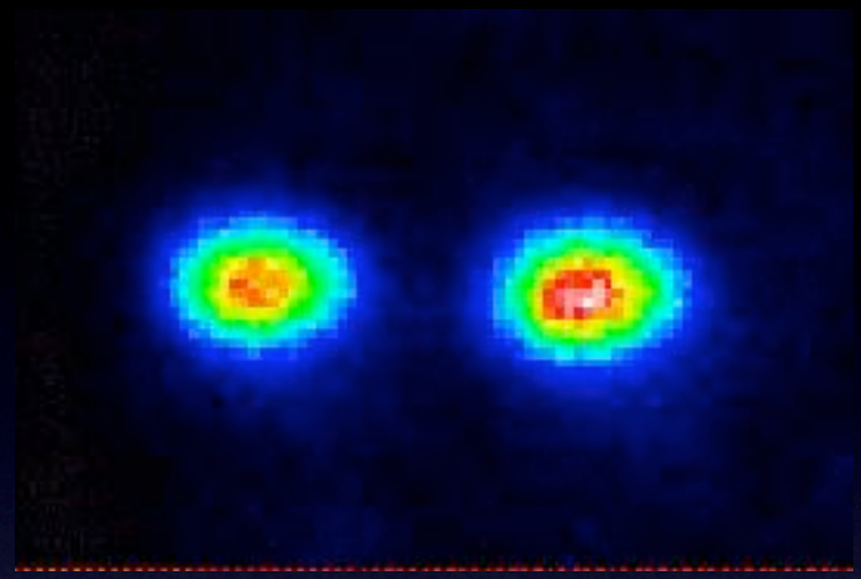


2 $^{9}Be^{+}$ Ions in an Ion Trap



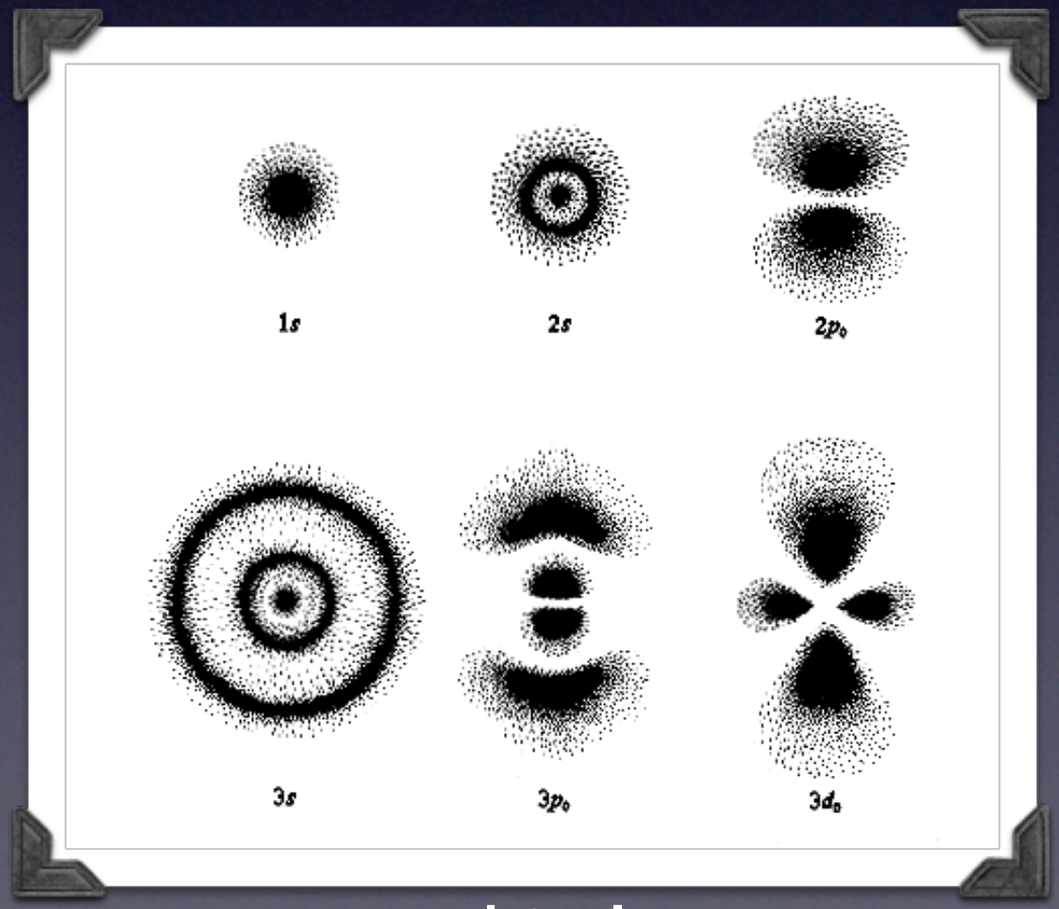
Where's the Qubit?





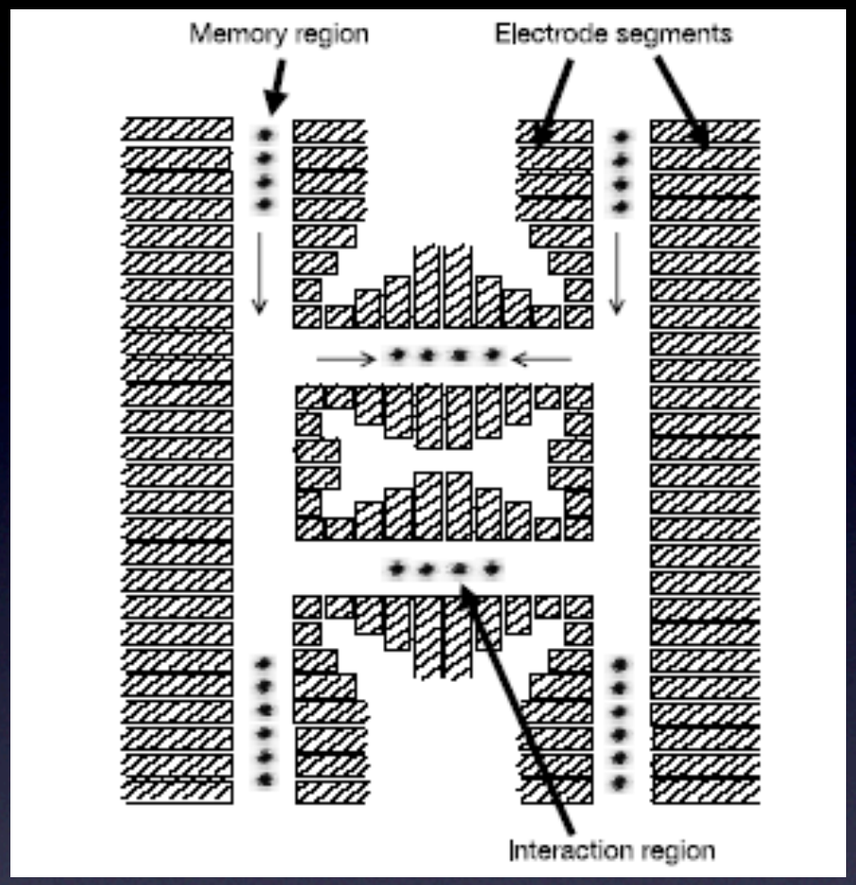
Each ion = 1 qubit

1. Well defined qubits



orbitals

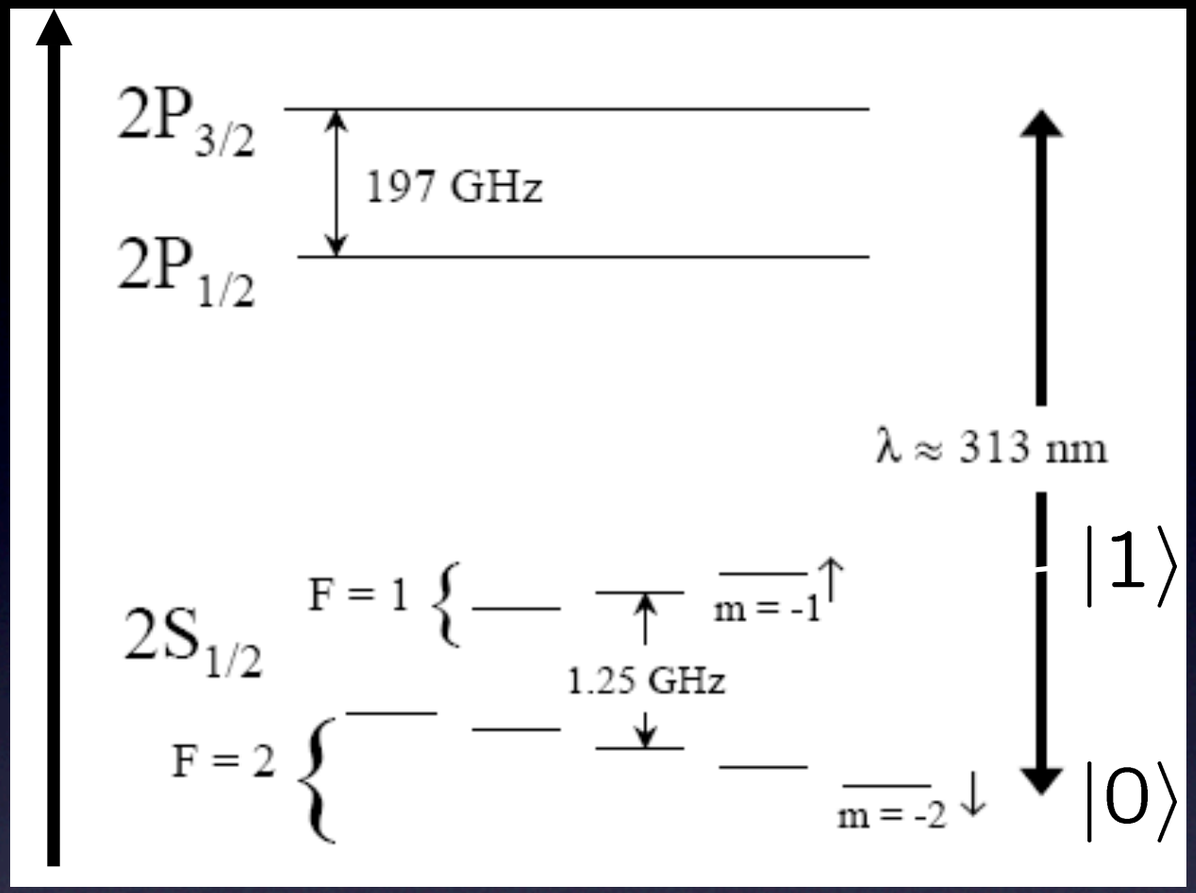
Scalable?

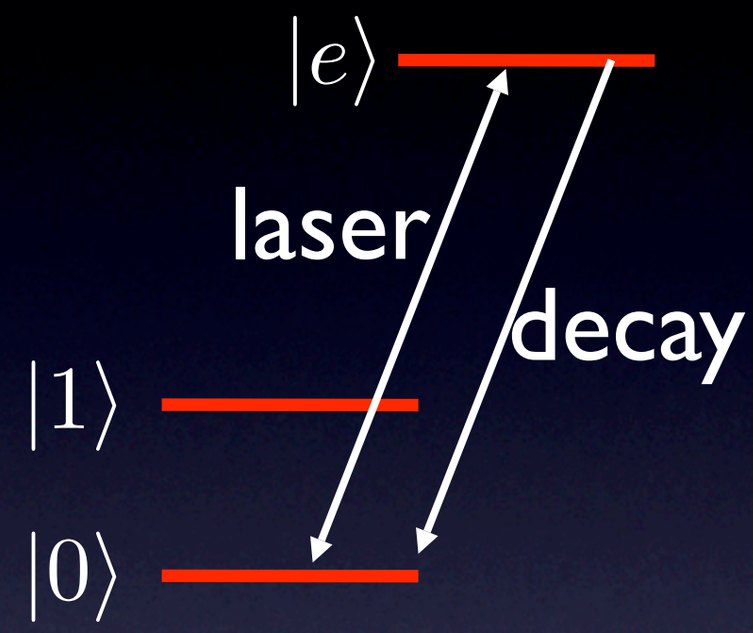


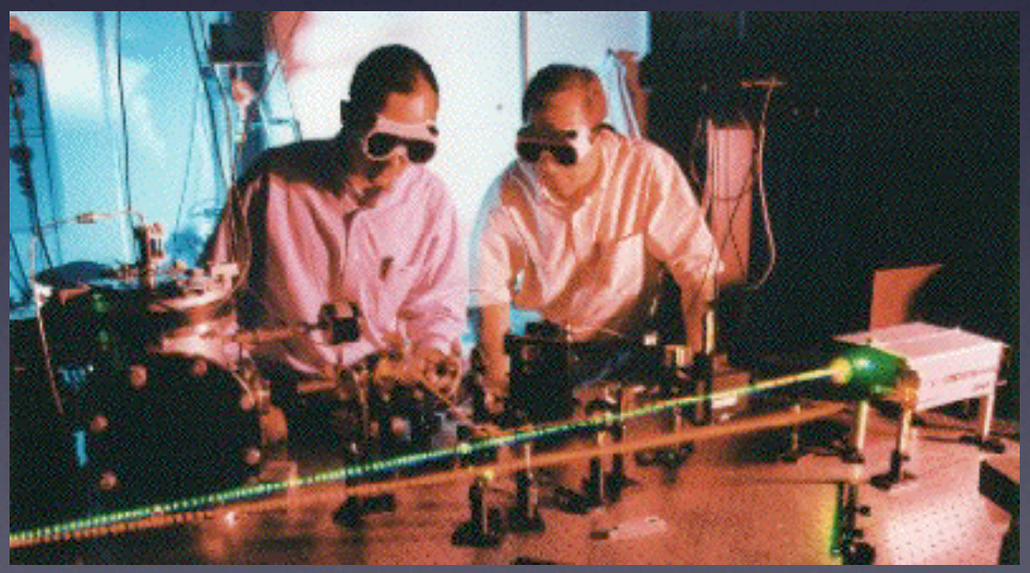
Well defined qubits in a scalable architecture?

Solid state qubits seem to have a huge advantage for scalability.

Measurement





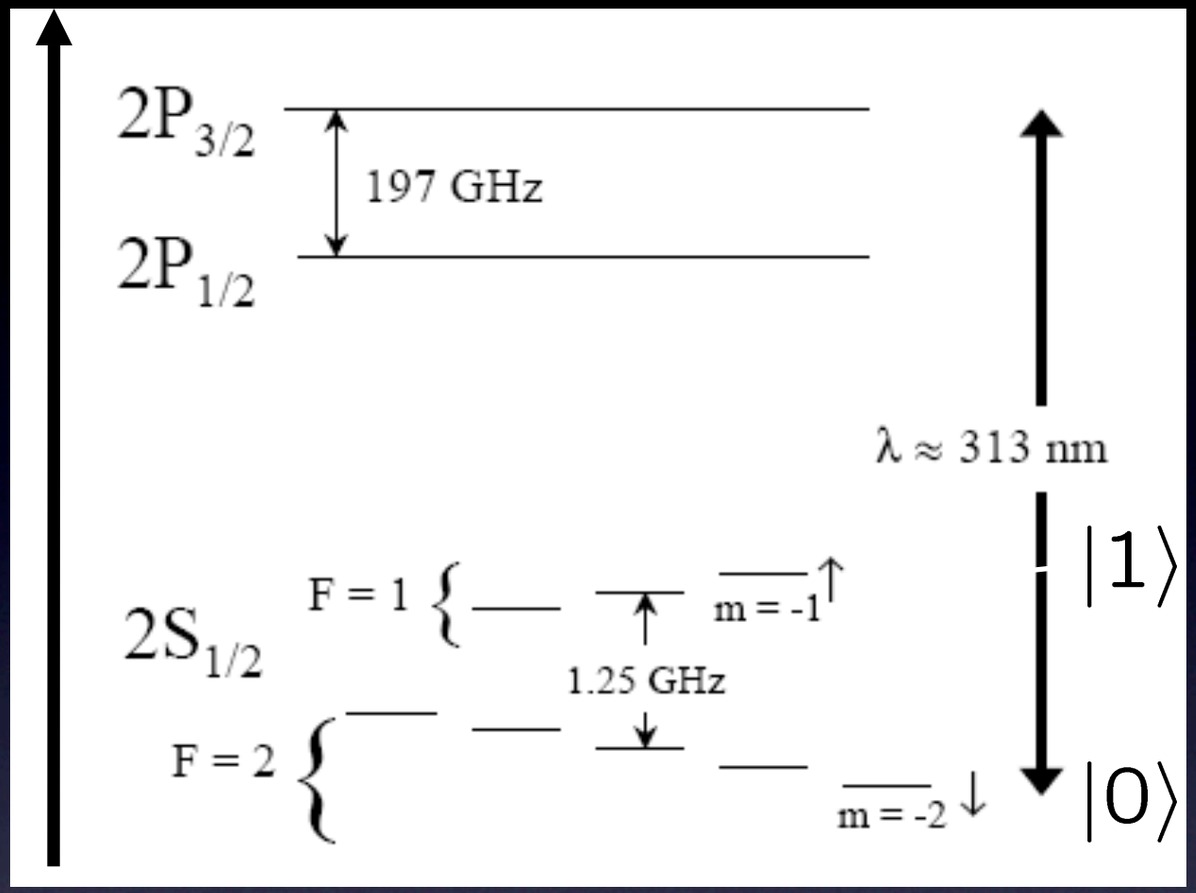


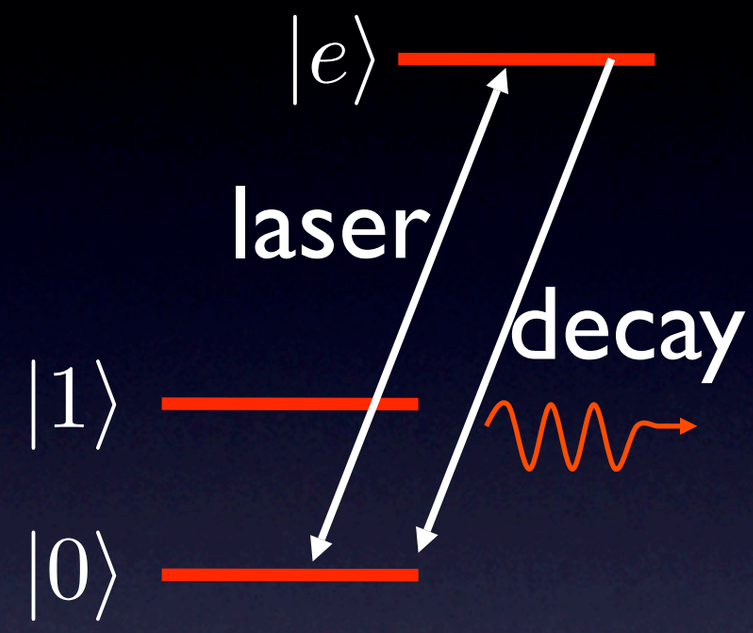
Detecting florescence implies in state 0

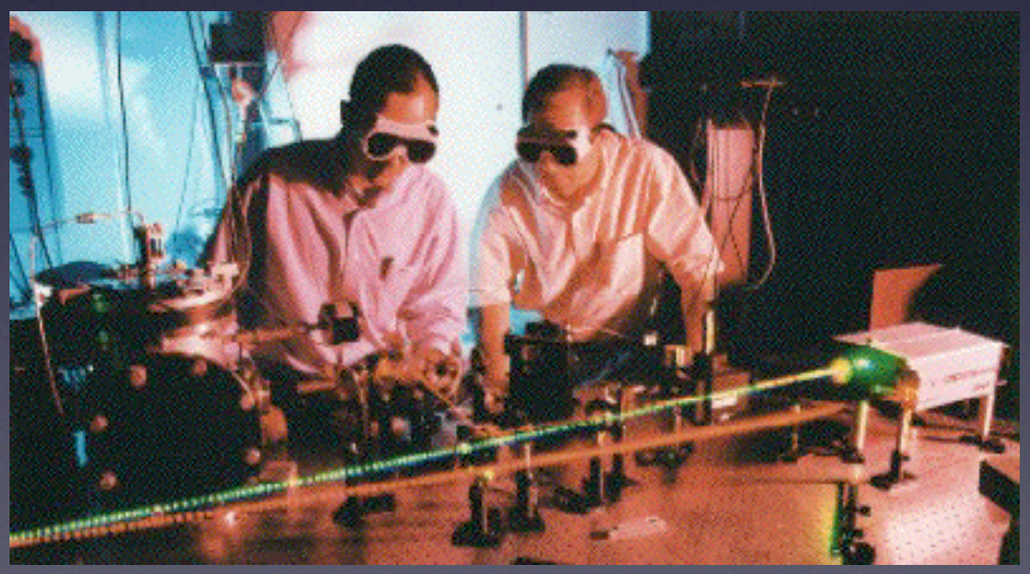
99.99% efficiency

5. Have the ability to perform high quality measurements

Measurement





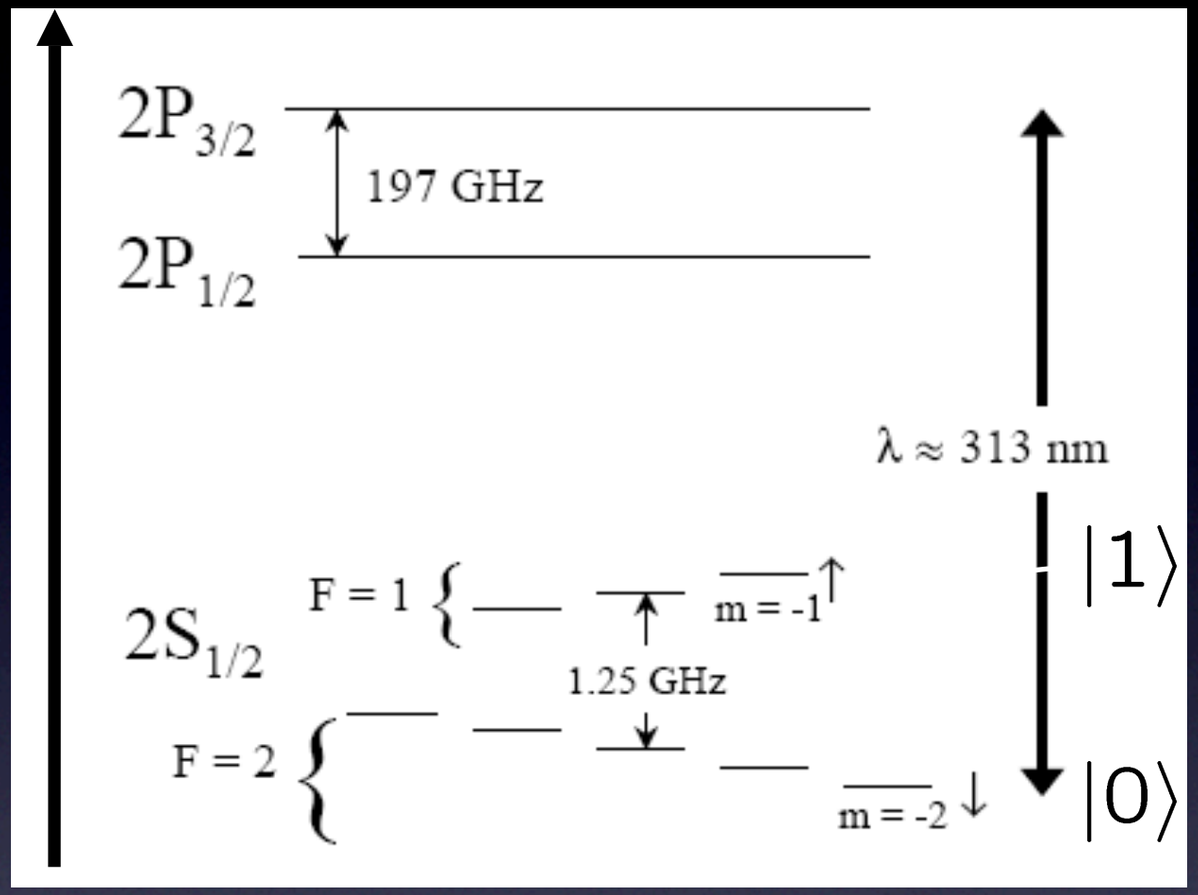


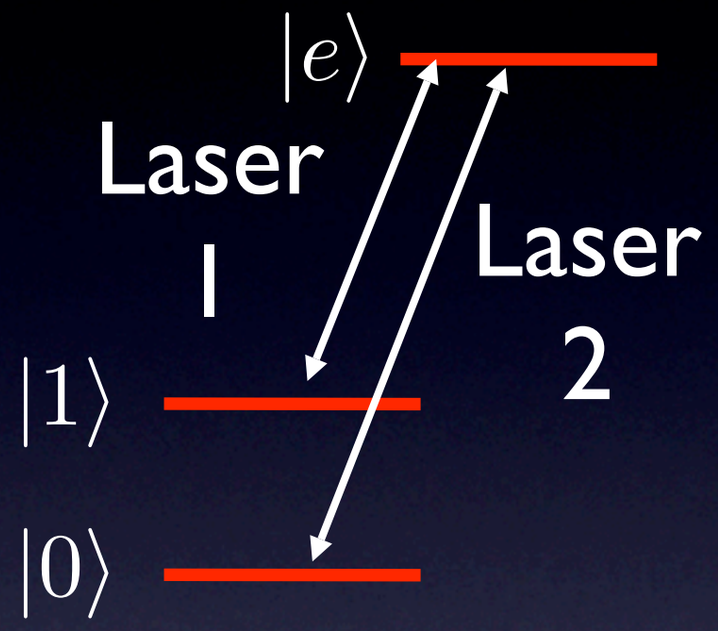
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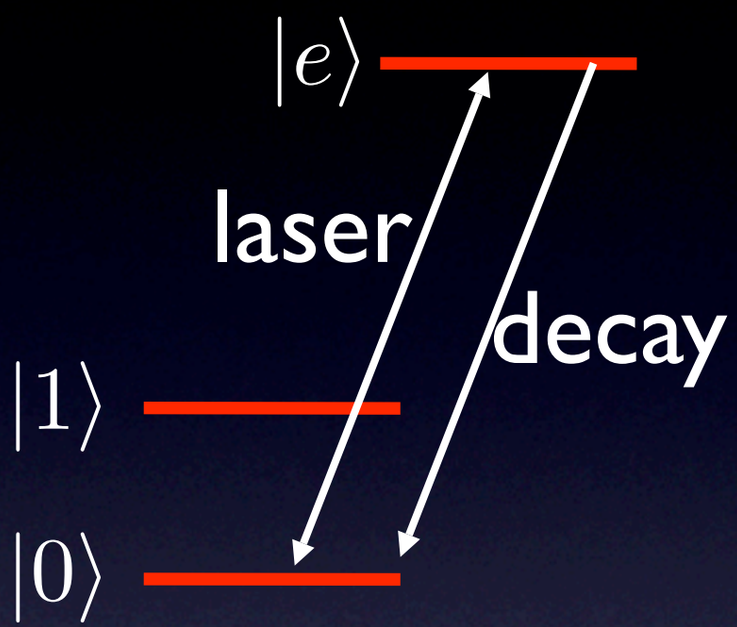
Single Qubit Operations

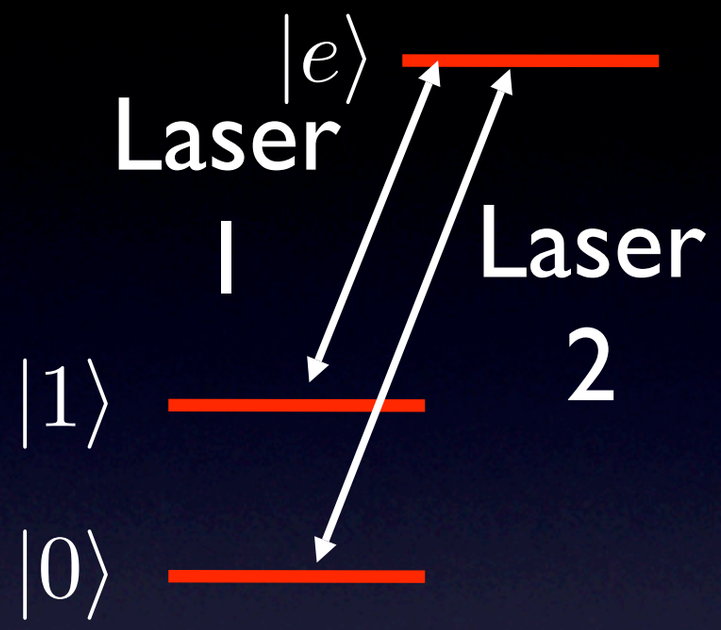




Allows any one qubit unitary operations $\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}$

Initialization



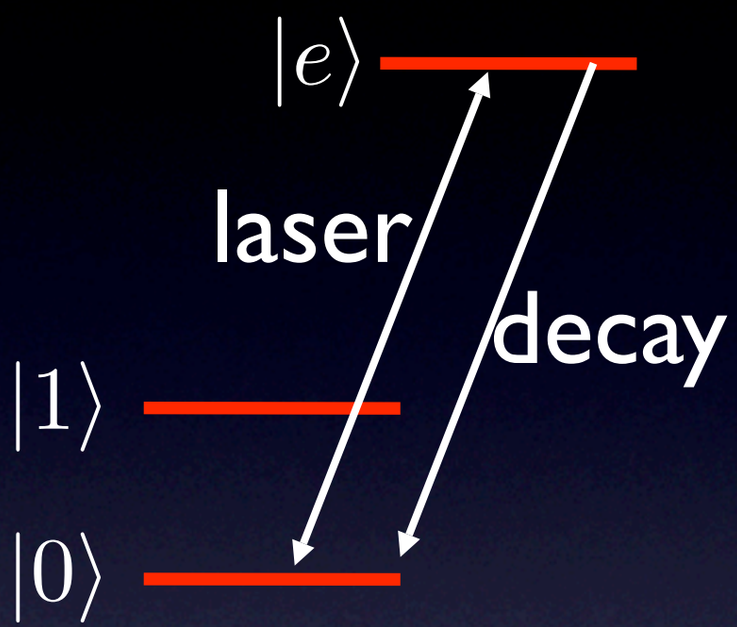


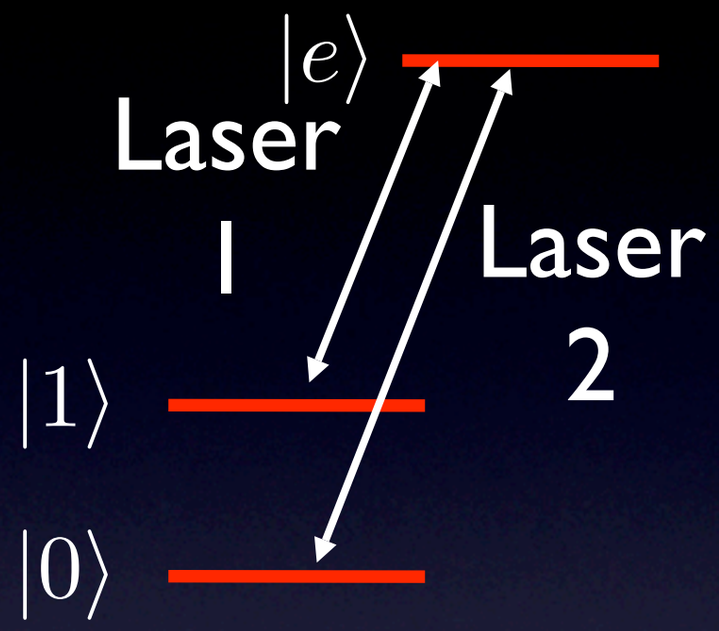
measure

If not in zero state, flip

2. The ability to initial the system to a deterministic state.

Initialization





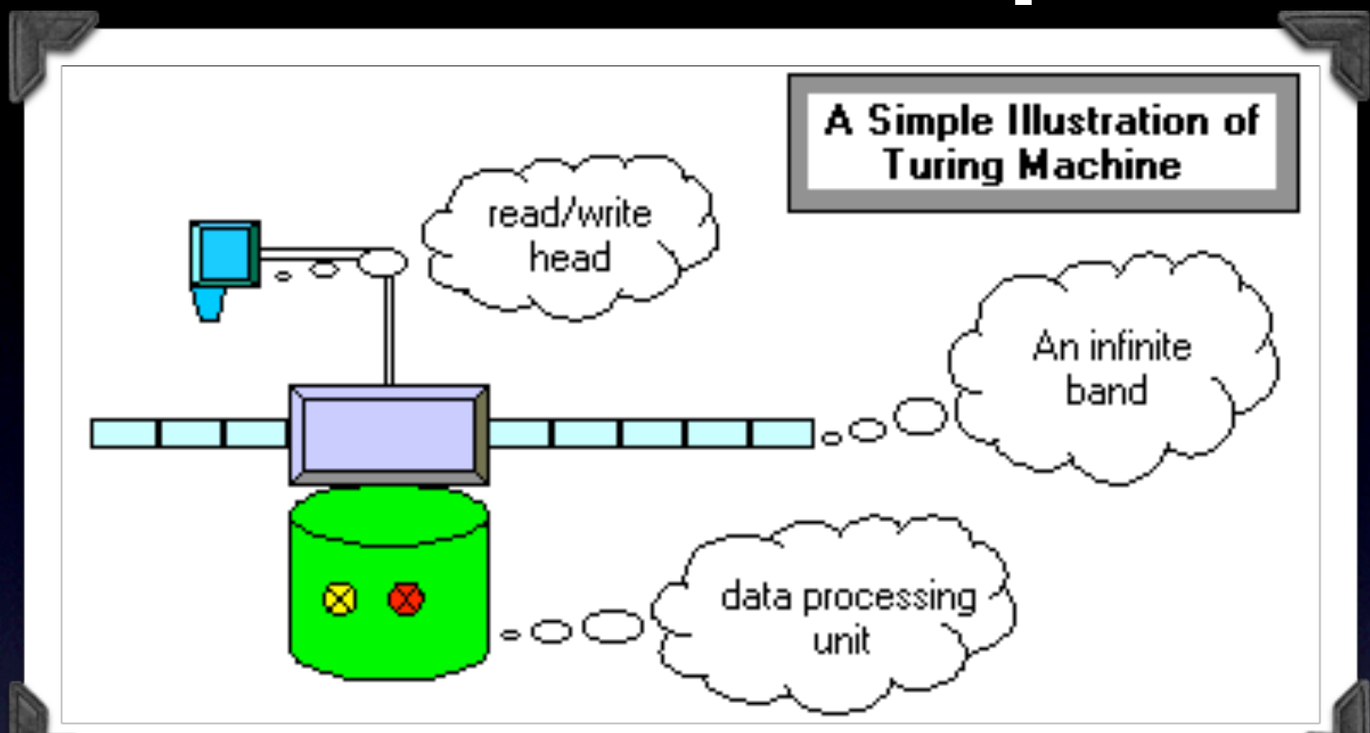
measure

If not in zero state, flip

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2. The ability to initial the system to a deterministic state.

Universal Computers



Turing machine reads state of tape at current position.

Based on this reading and state of machine, Turing machine writes new symbol at current position and possibly moves left or right, and updates state.

Certain Turing machines can perform certain tasks.

A Universal Turing Machine can act like any other possible Turing machine (i.e. it is programmable)

Universal Quantum Computer

Universal Quantum Computer

a quantum computer which can be programmed to perform any algorithmic manipulation on quantum information.

Set of Universal Quantum Gates

a set of operations/gates which, acting on the quantum information, can be used to implement (to any desired accuracy) any unitary evolution of the quantum info.

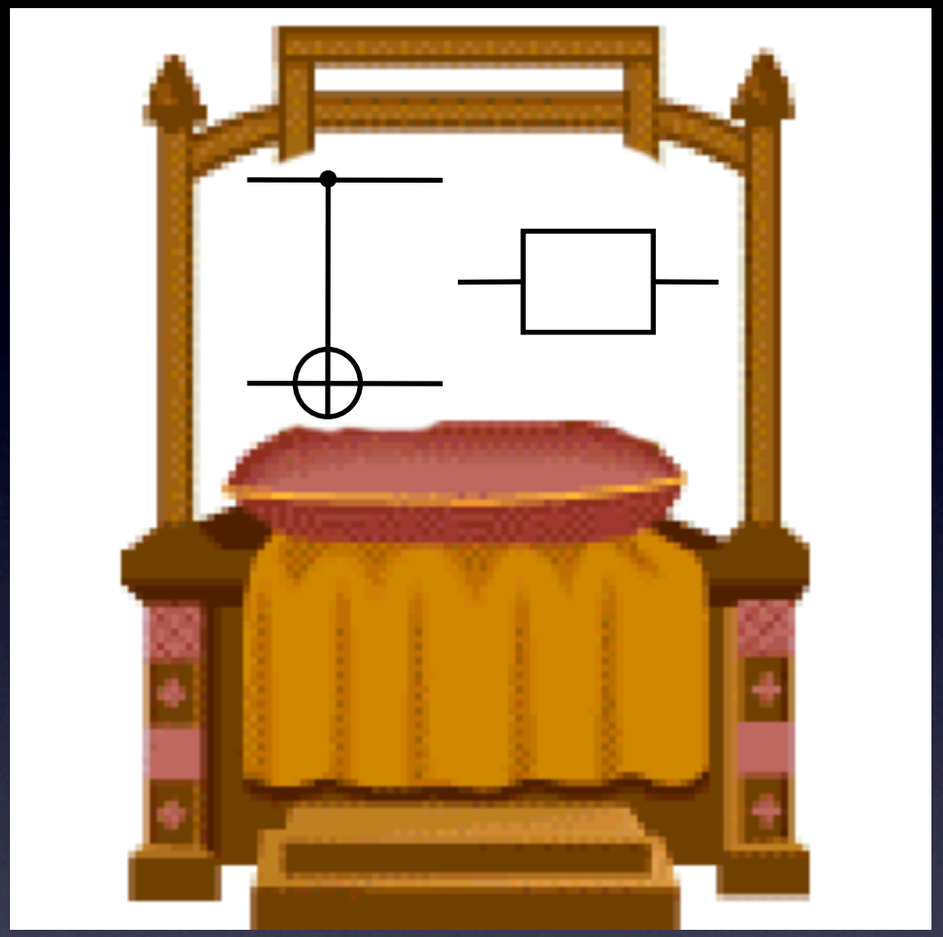
Universal Quantum Computer

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The Royal King and Queen of Universal Quantum Gates

CNOT and 1-qubit rotations

Coupling Two Qubits



stationary



sloshing mode

These modes can be used as a bus between the qubits.



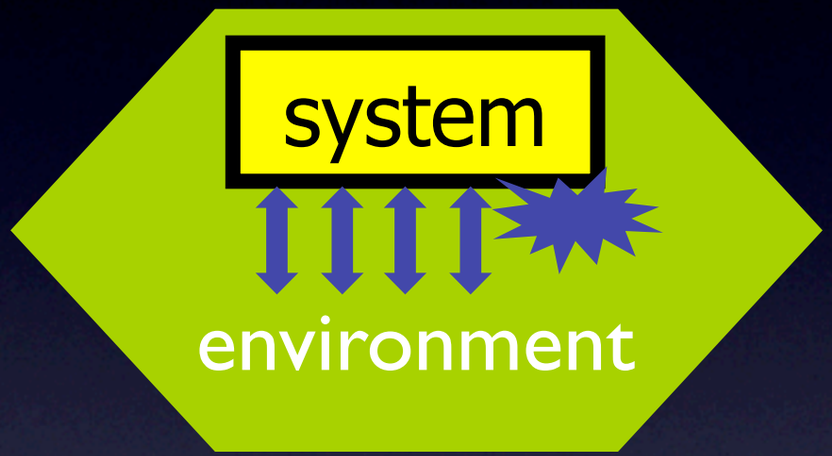
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

4. Have the ability to perform a universal set of quantum gates

What is the Problem?

3. Have faster control over the system than error processes in the system.

<u>Real quantum systems are open quantum systems!</u>



Quantum systems readily couple to an environment…

System decoheres: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ qubits → pbits **0**

$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix}$ 50% **0** 50% **1**

Quantum **Classical**

The Decoherence Problem (1996)

More Obstructions

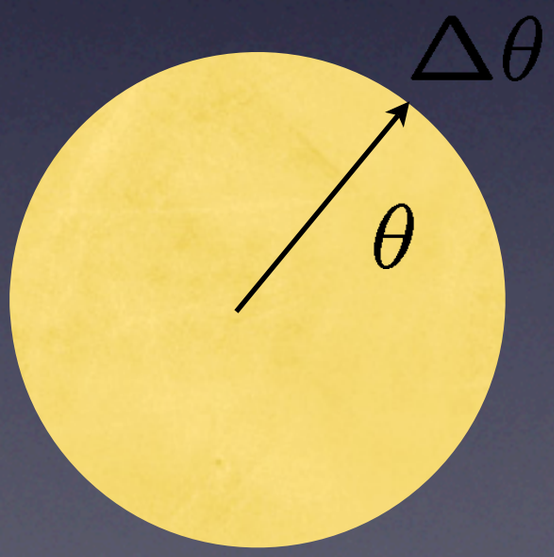
FAULTY CONTROL

System

Controller

Imprecise or faulty unitary gates



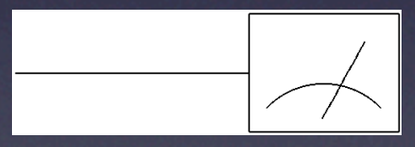


Imperfect state preparation



$$\rho \rightarrow (1-\epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$$

Imperfect measurements



$$Pr(0) = |\alpha|^2 - \epsilon$$

$$Pr(1) = |\beta|^2 + \epsilon$$

More Obstructions

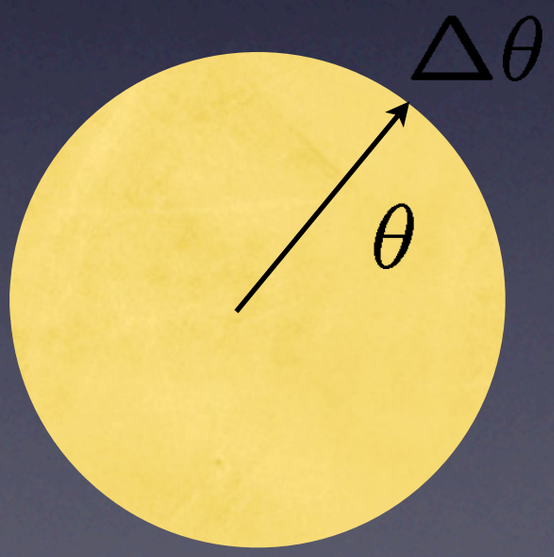
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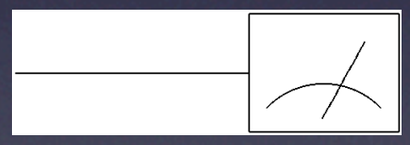


Imperfect state preparation



$$\rho \rightarrow (1-\epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$$

Imperfect measurements



$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$Pr(0) = |\alpha|^2 - \epsilon$$

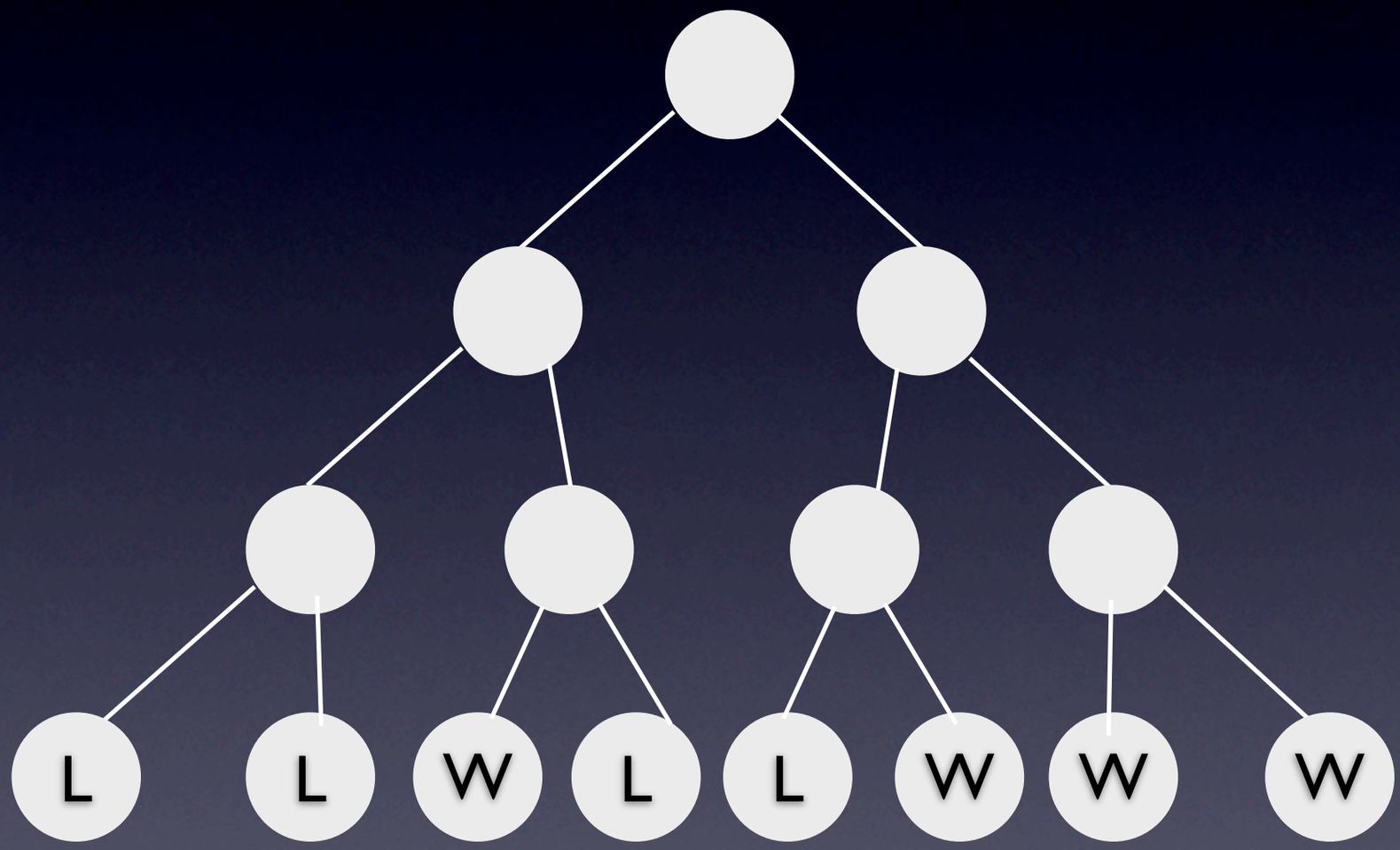
$$Pr(1) = |\beta|^2 + \epsilon$$

Games

player 1

player 2

player 1

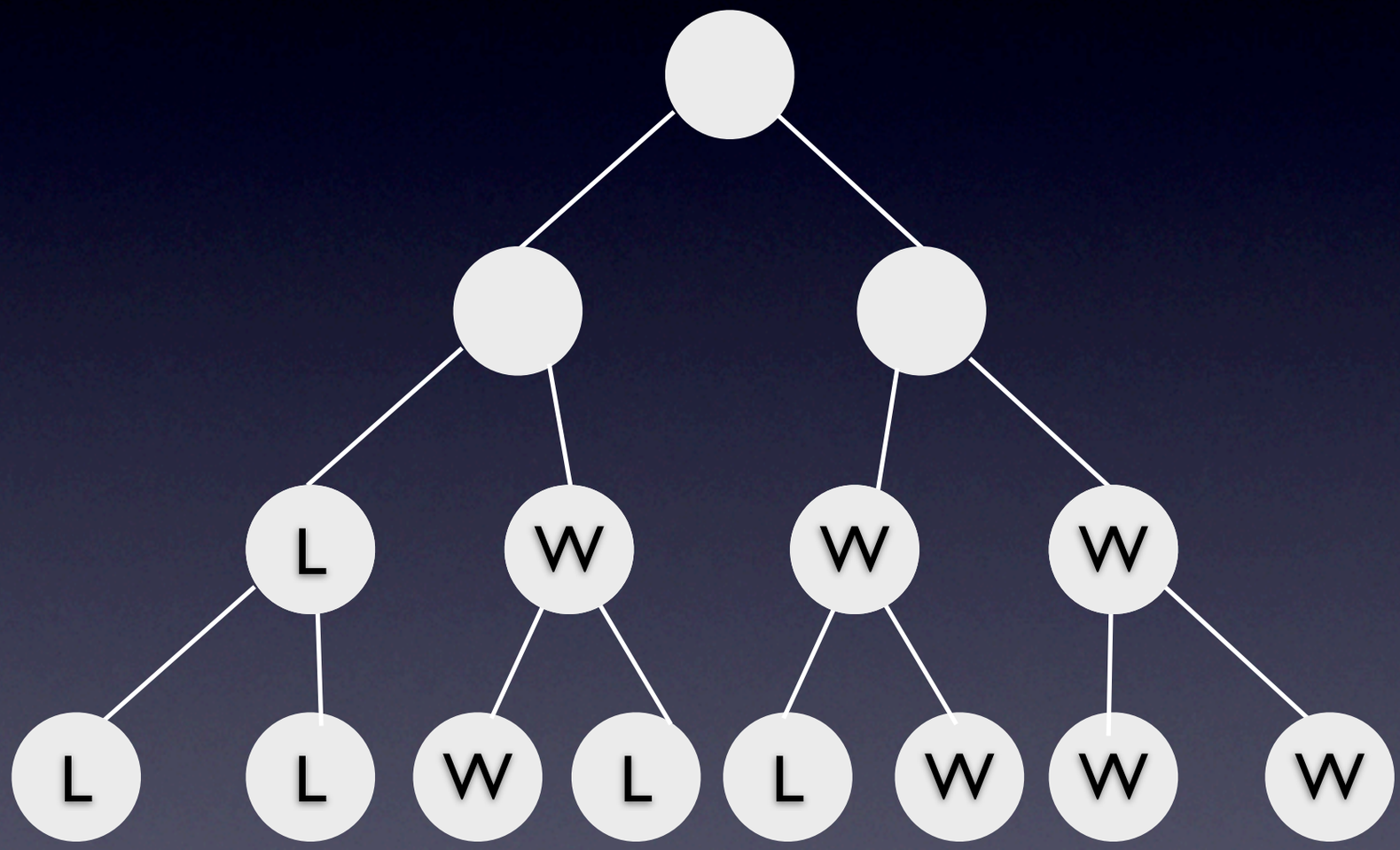


Games

player 1

player 2

player 1

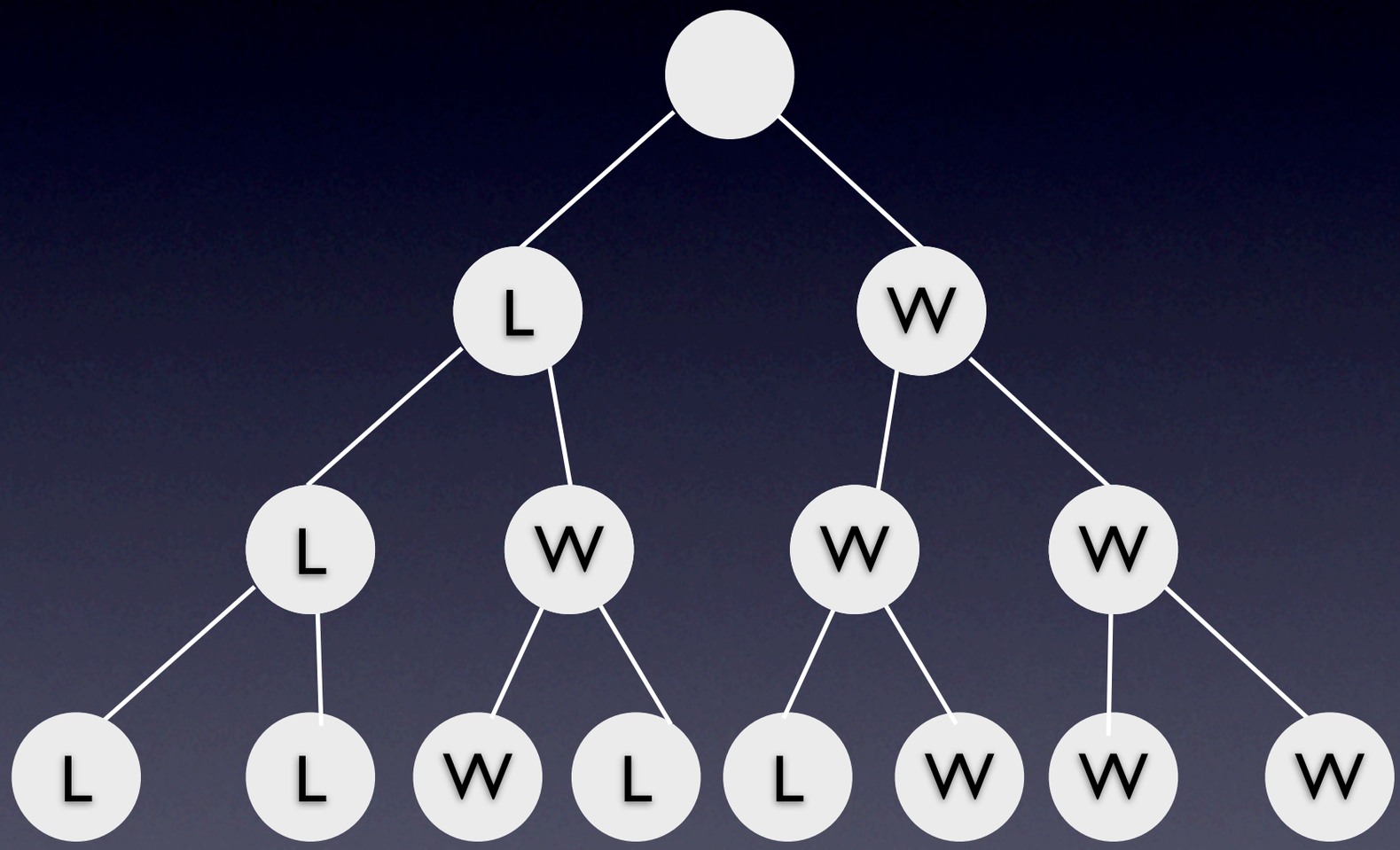


Games

player 1

player 2

player 1

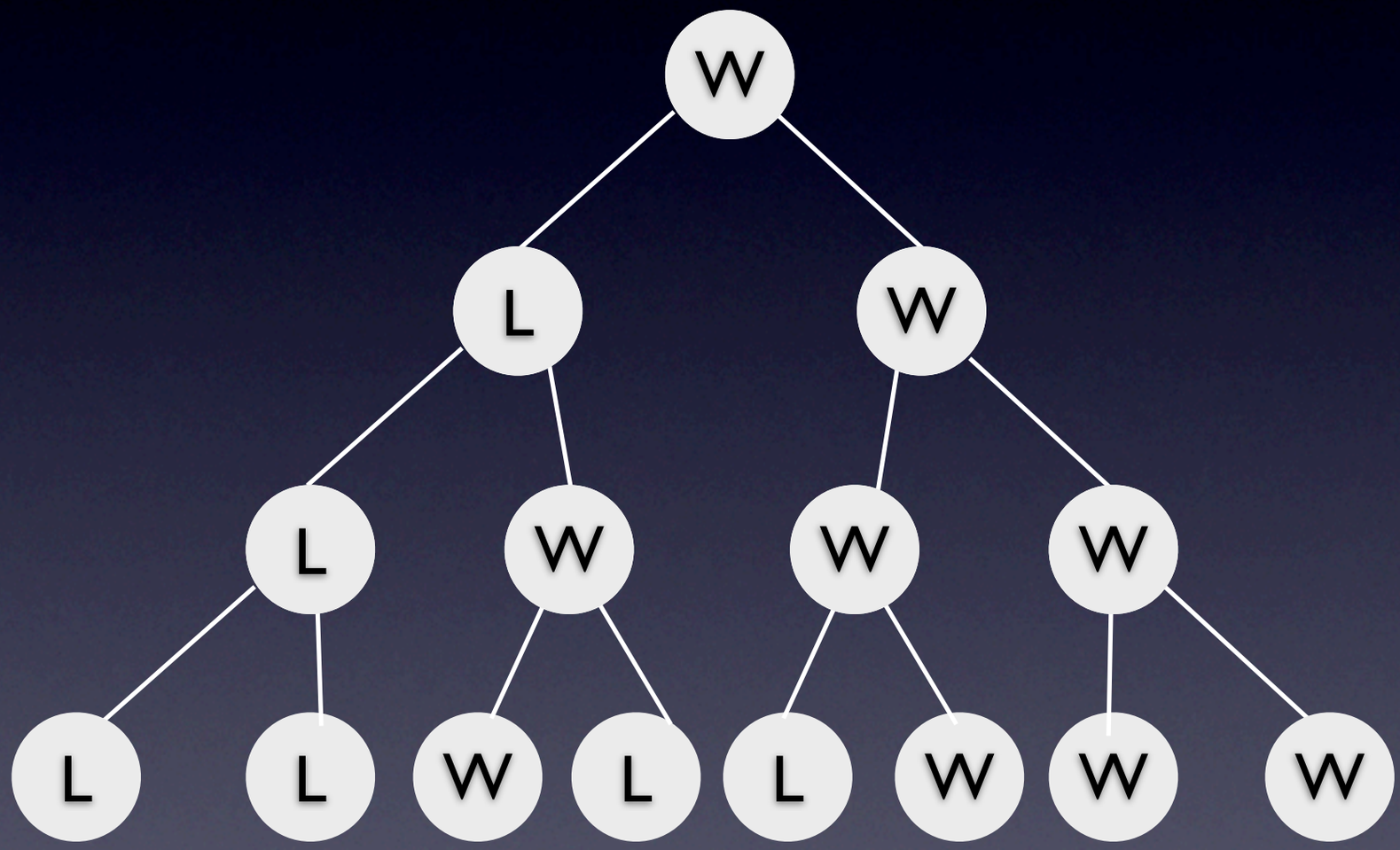


Games

player 1

player 2

player 1

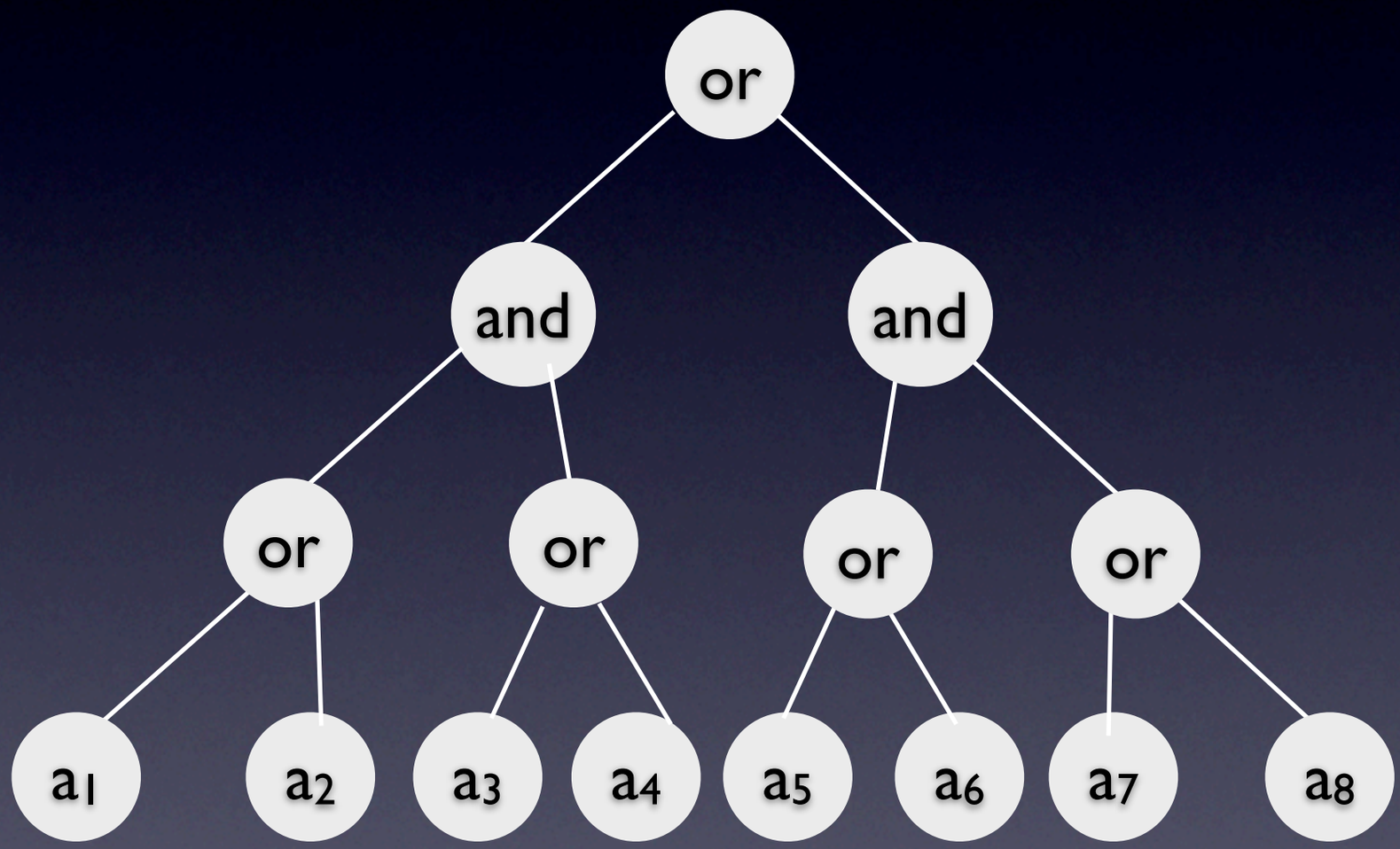


Games

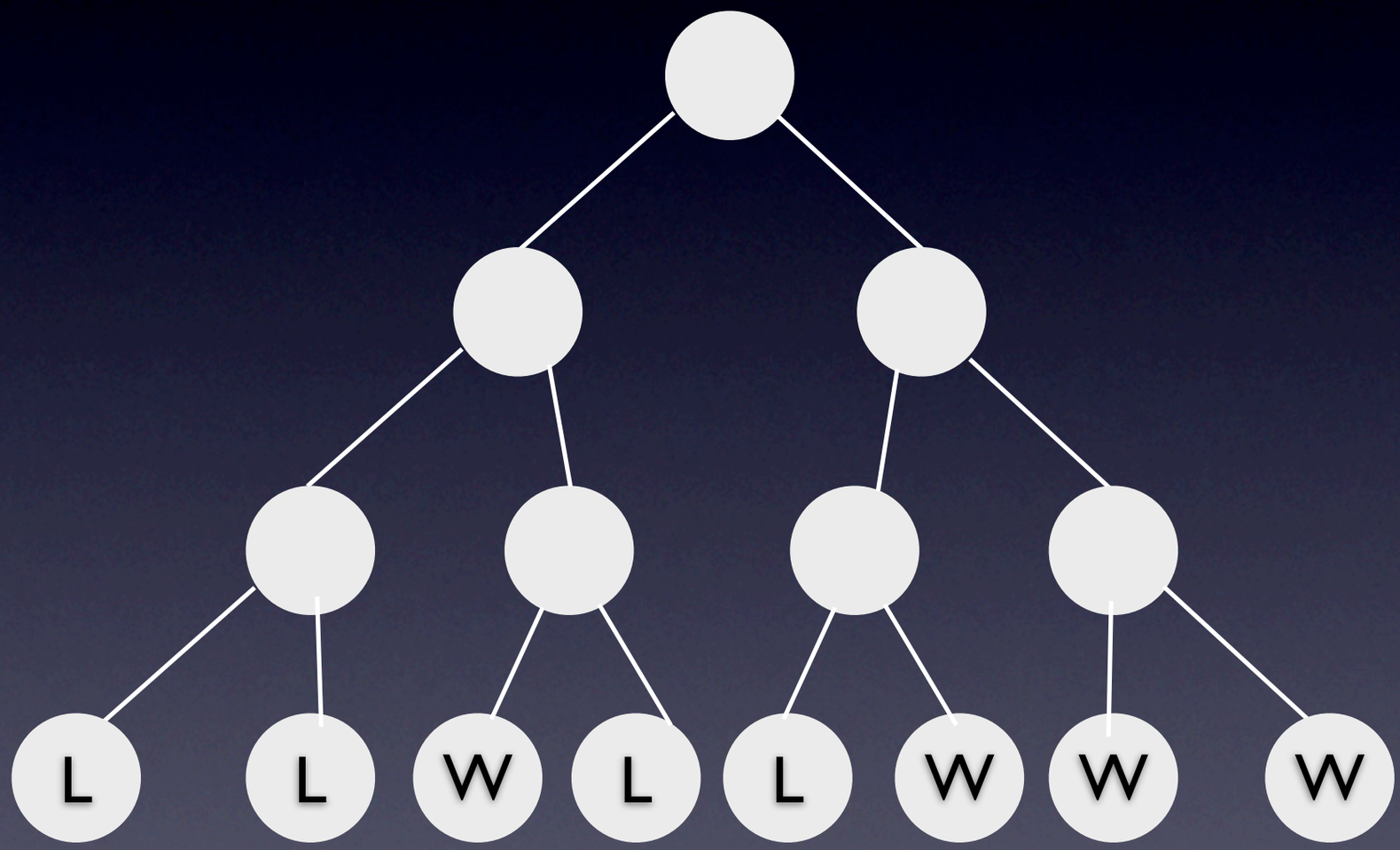
player 1

player 2

player 1

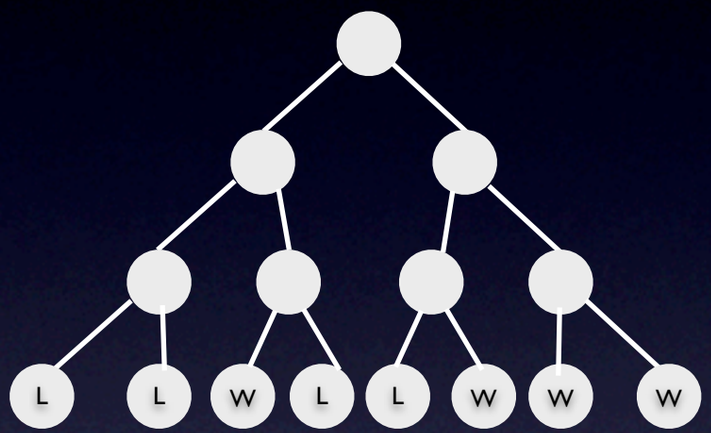


AND-OR Tree Evaluation



How many leaves do you need to query to compute value of root?

AND-OR Tree Evaluation



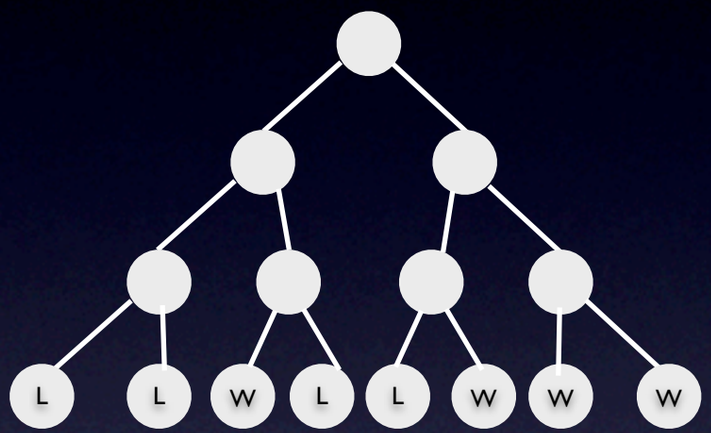
Classically (alpha-beta pruning)

$\Theta(N^\alpha)$

$\alpha = \log_2(1 + \sqrt{33}) - 2$

$\alpha \approx 0.75$

AND-OR Tree Evaluation



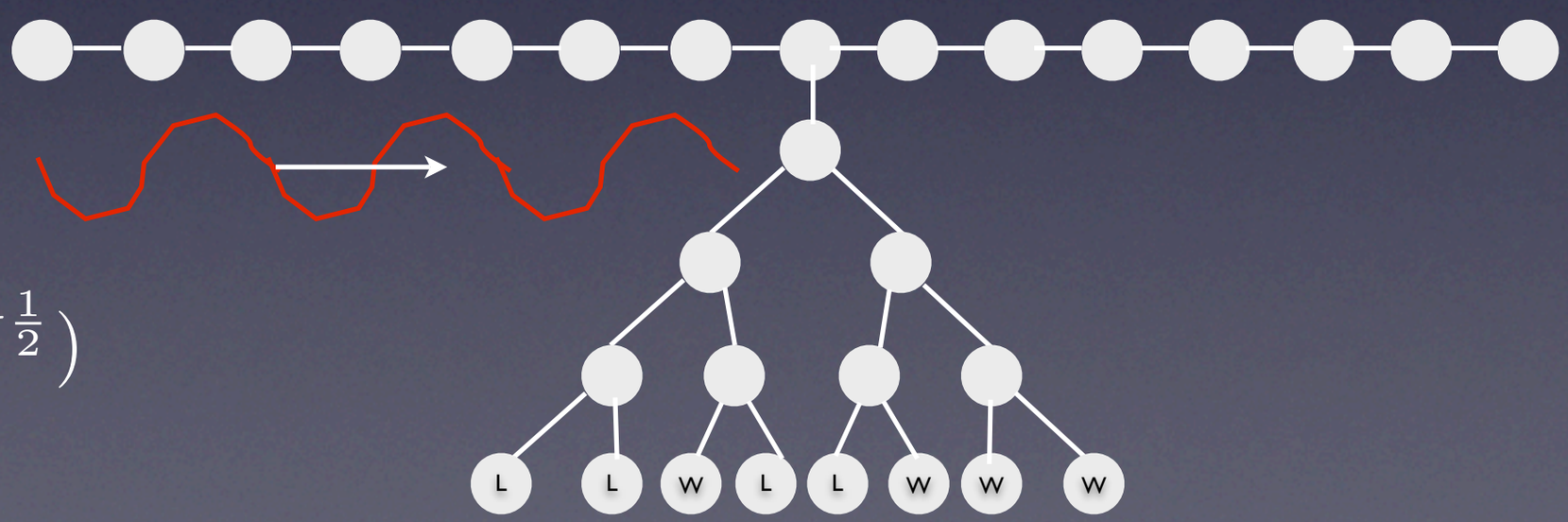
Classically (alpha-beta pruning)

$\Theta(N^{\alpha})$

$\alpha = \log_2(1+\sqrt{33}) - 2$

$\alpha \approx 0.75$

Quantum [Farhi, Goldstone, Gutman 2007]



$\Theta(N^{\frac{1}{2}})$